

BANGLAWILD: An In-the-Wild Bengali Scene Text Recognition Benchmark for OCR and Vision-Language Models

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Abstract

In-the-wild Bengali scene text recognition is largely unmeasured: existing resources target handwritten documents or constrained sign-board parsing, report only aggregate edit-distance metrics, and evaluate either conventional OCR or VLMs, never both on the same in-the-wild data. To address this gap, we introduce BANGLAWILD, a benchmark of 2,535 Bengali scene text images, each paired with a verbatim gold transcription, two categorical axes, four diagnostic attributes, and an orthographically standard form where the in-image text deviates from canonical spelling. We evaluate fifteen VLMs and three conventional OCR systems under three prompting strategies, fine-tune 6 open-source models with LoRA, and complement edit-distance metrics with an LLM-as-a-Judge evaluation. Our results reveal a persistent gap in which larger models within the same family do not outperform smaller ones. Our fifteen-class error taxonomy shows that visual mis-recognition accounts for ~60% of errors in the strongest systems, while conjunct-related errors contribute under 2%, challenging a long-standing assumption in Bengali OCR research; the same visual-dominant profile also holds across architectures, including the one conventional baseline that reads Bengali reliably. Prompt language mainly affects cross-script drift and LoRA reduces catastrophic failures in weak models without lifting the ceiling on already competent ones. Code and data will be publicly released.

1 Introduction

Vision-language models (VLMs) (Gemini Team, Google et al., 2025; Qwen Team, 2026) are advertised as broadly multilingual, yet the evidence supporting that claim skews sharply toward English and Chinese text, toward document OCR rather

than in-the-wild scene text, and toward short, clean image-text pairs rather than the cluttered surfaces on which text actually appears in public space (Fu et al., 2025; Tang et al., 2024). For low-resource Indic scripts, the gap between marketing and measurement is particularly acute. Although VLMs are routinely deployed in settings where Bengali scene text matters (transcribing addresses, receipts, election material, and medical packaging), we have no rigorous account of how, or how badly, they fail. Existing Bengali resources target conventional OCR settings such as cropped-word recognition or handwritten documents under controlled conditions (Rahman et al., 2022), leaving the perspective distortion, stylized typography, multilingual clutter, and occlusion that characterize Bengali text on public walls, street vehicles, banners, and product packaging entirely unmeasured. Crucially, no existing Bengali resource evaluates conventional OCR and generative VLMs on the same in-the-wild data, so it remains unknown whether modern VLMs improve over conventional OCR on this setting at all. We introduce BANGLAWILD, a diagnostic benchmark of 2,535 natively photographed Bengali scene text images captured on public-facing physical surfaces under uncontrolled conditions. Each image is paired with a verbatim gold transcription that preserves any in-image spelling errors or non-standard glyph substitutions, two categorical labels (CONTEXT, CONTENT), and four diagnostic attributes (CURVED TEXT, FONT STYLE, OCCLUSION, BACKGROUND COMPLEXITY). For the 3.43% of images whose in-image text deviates from standard spelling, an orthographically standard form is recorded in a separate field. This dual annotation lets us distinguish *recognition* errors (the model misreads what is present) from *correction* errors (the model silently rewrites a misspelling into its standard form), a distinction that, to our knowledge, no prior Bengali scene text benchmark supports. Our contributions are as fol-

*Equal contribution

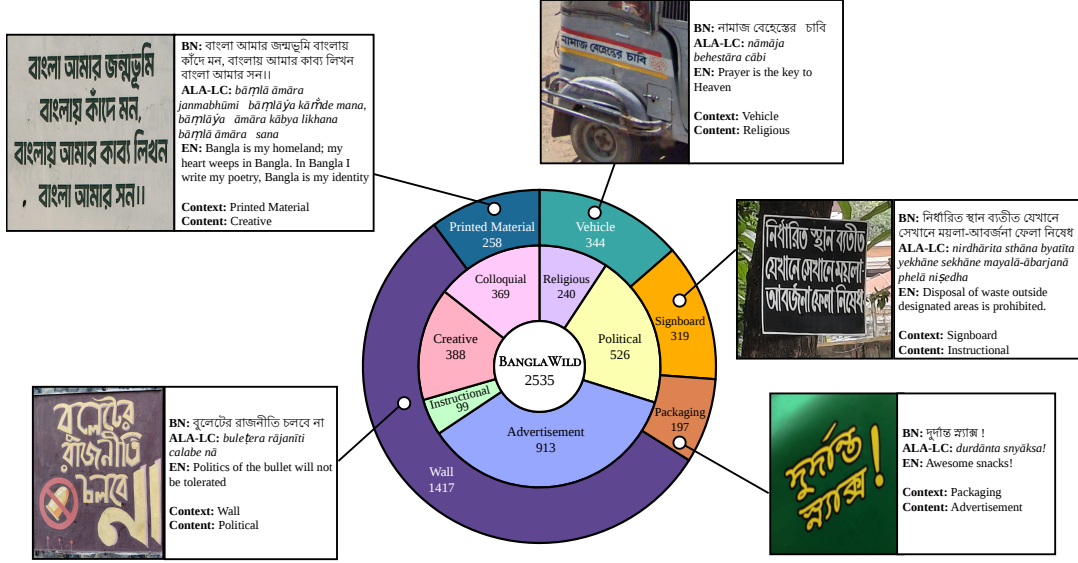


Figure 1: Category-wise distribution of BANGLAWILD across CONTEXT (outer ring) and CONTENT (inner ring) axes, with representative image-annotation pairs for each content category. Each annotation includes the original Bengali text (BN), an ALA-LC Romanization (American Library Association and Library of Congress, 2017) added here for readability, and an English translation (EN), neither of which is part of the released dataset.

lows:

- We present the first in-the-wild Bengali scene text recognition benchmark evaluating both conventional OCR and generative VLMs, stratified by categorical and diagnostic attributes with adjudicated gold transcriptions. Although the surface task is verbatim transcription, its dual verbatim/standard annotation targets two failure modes conventional OCR cannot structurally exhibit: silent over-correction, since a CRNN has no lexical prior to normalize a misspelling toward, and fluent hallucination toward plausible Bengali that a reader cannot flag from the output alone. We evaluate OCR alongside VLMs not because the task reduces to OCR, but to show the dominant failure mode is a property of the imagery, not one model class.
- We benchmark 15 contemporary VLMs and 3 conventional OCR systems across 3 prompting strategies and fine-tune 6 open-source models with LoRA, complemented by a validated LLM-as-a-Judge protocol on the full test set.
- We provide a fifteen-class error taxonomy that identifies visual mis-recognition, not orthographic confusion, as the dominant failure mode in Bengali scene text recognition, which holds across VLMs and a conventional OCR baseline, and bounds what prompt engineering and fine-tuning can achieve.

2 Related Work

Indic and Bengali Scene Text Recognition.

Multilingual scene text understanding for Indic scripts has been driven by the ICDAR Robust Reading competitions RRC-MLT-2017 and RRC-MLT-2019 (Nayef et al., 2019), which standardized multi-script benchmarks and evaluation protocols. Building on these, Gunna et al. (2022) investigate cross-script transfer learning from English to Hindi and Gujarati with CRNN and STAR-Net, evaluating transfer among six Indic languages, and show that cross-Indic transfer outperforms English-to-Indic transfer. For Bengali, public scene text resources remain scarce and target conventional OCR settings, including crowdsourced collection (Hossain and Rahman, 2023) and structured signboard parsing (Murad and Ali, 2023). Other Bengali resources target handwritten document recognition rather than scene text, including BN-HTRd (Rahman et al., 2022), the largest document-level Bengali HTR corpus with 108,147 annotated words across 788 pages, which remains the canonical reference for handwritten Bengali but is captured under controlled writing conditions. More recently, De et al. (2025) introduced BSTD, the largest multilingual Indic scene text benchmark to date with 6,582 images and 126,292 annotated words across 11 languages; only 6,304 (~5%) are Bengali, sourced from Wikimedia Commons rather than captured in-the-wild. These resources are built for CRNN or encoder-decoder pipelines

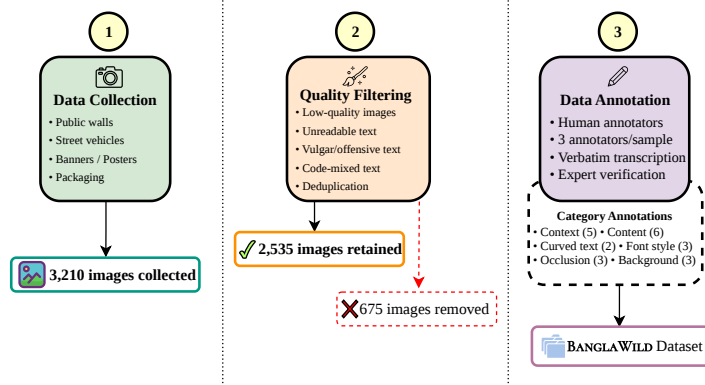


Figure 2: Overview of the BANGLAWILD Data Collection and Annotation Pipeline.

rather than generative VLMs, report only aggregate CER and WER, and do not isolate the script drift, conjunct degradation, or cross-script interference that uniquely affect generative VLMs on Indic scripts. We therefore do not claim the first Bengali scene text or OCR resource: Hossain and Rahman (2023) introduced a crowdsourced detection pipeline, and Murad and Ali (2023) proposed a structured signboard address parser, both valuable prior work. BANGLAWILD is instead the first in-the-wild Bengali scene text recognition benchmark to pair verbatim/standard dual annotation with evaluation of both conventional OCR and generative VLMs on the same data, none of which these resources provide.

Text-Centric and Indic VLM Benchmarks.

Text-centric VLM benchmarks have expanded evaluation beyond classical OCR, with OCRBench (Liu et al., 2024) first systematically evaluating multimodal models on OCR-centric tasks. OCRBench v2 (Fu et al., 2025) extended this with 10,000 human-verified QA pairs across 31 scenarios, exposing persistent weaknesses in localization and fine-grained perception. Both versions are English and Chinese focused (OCRBench v2 covers 31 scenarios but no Indic-script content) and thus do not surface the script-level failure modes relevant to Bengali. MTVQA (Tang et al., 2024) introduced the first multilingual text-centric VQA benchmark and identified the visual-textual misalignment problem from translated rather than visually grounded text, a constraint we adopt by sourcing only natively photographed Bengali scenes; MTVQA itself excludes Bengali and other Indic scripts. ThaiOCRBench (Nonesung et al., 2025) showed that low-resource scripts expose substantial hallucination and robustness failures in modern VLMs, directly motivating our cross-

script analysis. Sohail et al. (2024) report findings on 2,520 controlled images of Urdu, Albanian, and Tajik, showing that script complexity, not low-resource status alone, drives GPT-4o (OpenAI, 2024) failure. Concurrent work has identified the related failure mode of VLM *over-correction* in handwritten transcription (Seong et al., 2026), where models silently fix errors rather than transcribing them verbatim; our dual annotation extends this diagnostic capability to Bengali scene text, where orthographic non-standardness is the norm rather than the exception. Most relevant to our setting, IndicVisionBench (Faraz et al., 2025) provides the first large-scale Indic VLM benchmark, covering OCR, MMT, and VQA over English and 10 Indian languages through $\sim 5,000$ culturally grounded images and 37,000+ QA pairs. Its per-language OCR allocation is necessarily small once split across 11 languages and three tracks, and its imagery centers on cultural topics rather than uncontrolled urban scenes, so the benchmark emphasizes cultural understanding and cross-lingual evaluation over fine-grained script-level failure analysis on in-the-wild Bengali text.

3 The BANGLAWILD Dataset

We introduce BANGLAWILD, a benchmark of 2,535 in-the-wild images of Bengali scene text captured on public-facing physical surfaces under uncontrolled conditions. Each image includes a verbatim gold transcription, two categorical labels (CONTEXT, CONTENT), four diagnostic attributes (CURVED TEXT, FONT STYLE, OCCLUSION, BACKGROUND COMPLEXITY), and an orthographically standard form when the in-image text deviates from standard spelling. Details of the diagnostic attributes are provided in Appendix B. Fig. 2 summarizes the construction pipeline, and Fig. 1 reports

category-wise statistics.

Image Sourcing. Photographs were captured by trained collaborators using consumer mobile-phone cameras across five sources: *public walls, signboards, street vehicles, banners and posters, and consumer product packaging*. From an initial pool of **3,210** images, we discarded **675** (21.0%) that were (i) too low-quality or weathered for confident transcription, (ii) unreadable despite acceptable image quality, (iii) dominated by code-mixed Bengali-English text, or (iv) vulgar or offensive, yielding the final **2,535**-image benchmark. Filtering also included deduplication: near-identical images sharing both text and visual appearance were merged into a single representative, whereas identical text with different surfaces, fonts, layouts, or lighting was retained because it poses distinct recognition challenges. Incidentally captured faces and vehicle registration plates were blurred and verified as non-identifying.

Data Annotation. Each image is labeled along two orthogonal categorical axes. **CONTEXT** describes the physical medium on which text appears (*Vehicle, Signboard, Packaging, Wall, Printed Material*), while **CONTENT** captures its communicative intent (*Religious, Political, Advertisement, Instructional, Creative, Colloquial*). Formal definitions are provided in Appendix A.

We recruited **three** native Bengali-speaking annotators (university undergraduates with Bengali literature backgrounds, compensated at standard per-sample rates) to independently transcribe every image. Following the rubric (Appendix B), the gold transcription preserves the text *exactly as it appears*, including spelling errors and non-standard glyph substitutions. A separate field records the orthographically standard form for the **87** images (3.43%) containing non-standard spellings. This dual annotation distinguishes *recognition errors* (misreading the image) from *correction errors* (silently normalizing misspellings). For each of the 15 models under all 3 prompt variants, we classify outputs as matching the verbatim transcription, the standardized form, or neither (Tab. 15). Separately, two annotators labeled the **CONTEXT**, **CONTENT**, and four diagnostic attributes, with disagreements resolved by an expert in Bengali orthography. We use a 50/10/40 train/validation/test split (1,268/253/1,014 samples), evaluate all models on the common test split, and report results stratified

across all six label dimensions in Tabs. 11 and 12.

Annotation Verification. Disagreements among the three transcribers were resolved by the expert verifier to produce the canonical gold transcription. Overall, 18.4% of images required adjudication on at least one character; the remainder achieved unanimous first-pass agreement. Mean pairwise CER across all 2,535 images is 3.2%, and Cohen’s κ ranges from 0.81–0.93 across all categorical labels, indicating strong agreement (full breakdown in Tab. 4).

4 Experiment Design

We benchmark 15 vision-language models (full list in Tab. 7) under two regimes, (i) zero-shot prompting on the full BANGLA WILD test set and (ii) LoRA fine-tuning of the 6 open-source models with publicly available weights, and additionally evaluate 3 conventional OCR systems as non-generative baselines, scored with the same pipeline as the VLM outputs. We report CER, WER, and $1 - \text{NED}$ in Tab. 1, and additionally an LLM-as-a-Judge score (Zheng et al., 2023) on a 1-5 rubric-anchored scale to complement edit-distance metrics with a meaning-aware signal; the judge prompt and validation protocol are in Appendix F.

4.1 Zero-Shot Prompting

We evaluate using three prompting strategies: English and Bengali baseline prompts, and a structured metadata prompt. P1 and P2 are image-only conditions and constitute our deployment-realistic setting. P3 additionally prepends the gold **CONTEXT** and **CONTENT** labels, which are annotator-provided and unavailable at inference time; we therefore treat P3 as an *oracle condition* that upper-bounds the value of perfect scene priors, report it for reference only, and base all headline claims and best-prompt comparisons on {P1, P2}. Prompting details and examples are in Appendix E.

4.2 Parameter-Efficient Fine-Tuning

We apply LoRA (Hu et al., 2022) to test whether targeted adaptation improves Bengali scene text recognition beyond zero-shot prompting. For a pre-trained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$, LoRA freezes $\mathbf{W}_0$ and learns a low-rank update

$$\mathbf{W}_0 + \Delta\mathbf{W} = \mathbf{W}_0 + \mathbf{B}\mathbf{A},$$

where $\mathbf{B} \in \mathbb{R}^{d \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. Only $\mathbf{A}$ and $\mathbf{B}$ receive gradient updates. At in-

ference we use the merged weights $\mathbf{W} = \mathbf{W}_0 + \mathbf{BA}$, incurring no additional latency. Each model is trained on (image, prompt) pairs against the ground-truth transcription using only P1. LoRA adapters are applied to the attention projections (q_proj , k_proj , v_proj , o_proj) and MLP projections (up_proj , $down_proj$, $gate_proj$) of the *language-model component only*; the vision encoder and any vision-language connector are kept frozen throughout training. Training settings are detailed in Appendix G.2.

5 Result Analysis

A persistent gap despite larger models. Tab. 1 shows that the best system on BANGLAWILD, Gemini 2.5 Flash, achieves **CER 14.08** and WER 29.97 under P1, its best deployment-realistic prompt (P3 reaches CER 13.93 but is an oracle condition and excluded from headline comparisons; see §4.1). Performance is not monotonic with model size: Gemini 2.5 Flash outperforms the larger Gemini 2.5 Pro (CER 14.08 vs. 21.65; WER 29.97 vs. 36.42), while within Qwen3.5, the 9B model slightly surpasses the 27B dense and 35B-A3B MoE variants. As the 9B dense, 27B dense, and 35B-A3B MoE models differ architecturally rather than forming a clean scaling series, this non-monotonicity should be interpreted cautiously. InternVL3.5 and GPT-5.4-nano collapse across scales (CER >100), often producing outputs longer than the ground truth. These flat or inverse scaling trends suggest that Bengali scene text recognition on BANGLAWILD depends more on training data composition than model size.

Prompt language has a small but systematic effect on the mid-tier. Bengali-instructed P2 improves over English P1 on CER for 7 of 15 models, with the largest gains in the mid-tier (Claude Sonnet 4.6: $\Delta\text{CER} = -4.69$; Claude Haiku 4.5: -8.30 ; Gemini 2.5 Pro: -3.20 ; GPT-5.4-nano: -54.15). The effect reverses for the weakest models: InternVL3.5-4B degrades from 118.44 to 156.92 and InternVL3.5-2B from 132.73 to 176.70 under P2, with WER showing the same trend, suggesting these models transcribe the Bengali instruction instead of treating it as a directive, thereby lengthening outputs and inflating both metrics. P3, the oracle metadata condition, achieves the best CER for 9 of 15 models (and every aggregate metric for the weakest open models) but rarely the best 1-NED, indicating that gold scene priors mainly

help models commit to a transcription target rather than improve semantic fidelity. Because this metadata is unavailable at inference, P3 is reported for reference only.

Fine-tuning rescues catastrophic failures but does not raise the ceiling. LoRA on the 50% training split yields large CER reductions for the weakest open-source models: Gemma-3-4B-it drops from 134.82 $\rightarrow$ 50.72 (-62% relative), InternVL3.5-8B from 116.37 $\rightarrow$ 45.81 (-61%), and the two smallest InternVL variants from 118–133 to the low 50s (-56% , -61%). By contrast, the strongest open model, Qwen3.5-9B, slightly regresses (20.38 $\rightarrow$ 21.53), while Gemma-3-12B-it improves on CER (63.89 $\rightarrow$ 28.91) and 1-NED (50.47 $\rightarrow$ 72.79) but still trails zero-shot Qwen3.5-9B. The key finding: LoRA on 1,268 images mitigates catastrophic failures (refusals, off-target English captions, and cross-script contamination) rather than improving the Bengali recognition already achieved by competent models.

Edit-distance and semantic judgments diverge at the top. At the top of the leaderboard, CER and the LLM judge disagree: Claude Sonnet 4.6 achieves the highest judge score across all fifteen systems (3.82 at P2) despite trailing Gemini 2.5 Flash on CER by about four points (18.46 vs. 14.08 at each model’s best deployment-realistic prompt). Gemini 2.5 Pro (judge 3.67) ranks between Flash (3.69) and Qwen3.5-9B (3.30), again differing from the CER ordering. As discussed in §6, Sonnet appears to produce more spelling-normalized yet meaning-preserving transcriptions, which edit distance penalizes but the judge does not. So, we treat the judge score as complementary to CER rather than a standalone ranking metric.

Cross-family judge validation isolates in-family bias. The primary judge in Tab. 1 is Gemini 2.5 Pro, which evaluates all fifteen systems under the rubric in Appendix F. Because Gemini 2.5 Flash and Gemini 2.5 Pro are judged by a model from the same family, prior work on self-preference, family bias, and preference leakage (Panickssery et al., 2024; Wataoka et al., 2024; Spiliopoulou et al., 2025; Li et al., 2025) suggests their scores may be inflated. To quantify this, we re-scored both Gemini models with Claude Sonnet 4.6 using the identical rubric. Gemini 2.5 Flash drops from {3.84, 3.86, 3.86} (Gemini Pro judge; P1/P2/P3) to {3.68, 3.69, 3.70} (Sonnet judge),

Model	CER↓			WER↓			1-NED↑			Judge↑		
	P1	P2	P3†	P1	P2	P3†	P1	P2	P3†	P1	P2	P3†
<i>Zero-Shot Prompting</i>												
<i>Closed-source VLMs</i>												
Gemini 2.5 Flash	14.08 [12.62,15.54]	15.60 [13.69,17.51]	13.93 [12.63,15.23]	29.97 [28.08,31.86]	31.37 [29.11,33.63]	30.22 [28.39,32.05]	82.11 [81.13,83.09]	82.55 [81.61,83.49]	82.25 [81.28,83.22]	3.68*	3.69*	3.70*
Claude Sonnet 4.6	23.15 [17.02,29.28]	18.46 [13.12,23.80]	17.96 [14.47,21.45]	43.58 [37.77,49.39]	39.67 [34.89,44.55]	38.50 [35.18,41.82]	78.89 [77.81,79.97]	80.12 [79.10,81.14]	79.47 [78.40,80.54]	3.78	3.82	3.81
Gemini 2.5 Pro	24.85 [23.26,26.44]	21.65 [20.11,23.19]	23.95 [22.31,25.59]	39.53 [37.50,41.56]	36.42 [34.52,38.32]	38.29 [36.32,40.26]	76.10 [74.87,77.33]	78.32 [77.12,79.52]	76.94 [75.69,78.19]	3.58*	3.67*	3.63*
Llama 4 Maverick	36.96 [33.20,40.72]	37.98 [34.20,41.76]	37.74 [33.9,41.6]	59.43 [53.19,65.67]	60.60 [54.2,67.1]	59.89 [52.95,66.83]	58.69 [57.33,60.05]	60.61 [59.34,61.88]	58.73 [57.37,60.09]	2.65	2.66	2.65
GPT-5.4-mini	53.41 [50.84,55.98]	56.83 [52.54,61.12]	52.49 [49.65,55.33]	81.17 [78.90,83.44]	85.14 [80.96,89.32]	80.11 [77.22,83.00]	42.70 [41.45,43.95]	44.19 [43.04,45.34]	43.03 [41.81,44.25]	1.77	1.77	1.80
Claude Haiku 4.5	90.47 [66.17,114.77]	82.17 [66.89,97.45]	88.72 [70.40,107.04]	113.33 [94.03,132.63]	108.68 [93.26,124.10]	113.45 [95.53,131.37]	28.44 [27.36,29.52]	32.09 [31.12,33.06]	28.45 [27.36,29.54]	2.15	2.25	2.22
GPT-5.4-nano	246.95 [218.4,280.7]	192.80 [170.2,220.8]	114.82 [105.5,125.1]	349.15 [303.74,394.56]	236.03 [205.88,266.18]	140.50 [131.18,149.82]	11.53 [10.94,12.12]	15.25 [14.70,15.80]	11.43 [10.96,11.90]	1.02	1.08	1.09
<i>Open-source VLMs</i>												
Qwen3.5-9B	20.38 [19.97,20.79]	20.51 [17.36,23.66]	20.04 [18.65,21.43]	45.23 [43.52,46.94]	45.90 [43.12,48.68]	44.91 [43.10,46.72]	74.68 [73.67,75.69]	75.57 [74.61,76.53]	74.83 [73.82,75.84]	3.29	3.30	3.30
Qwen3.5-35B-A3B	21.20 [20.89,21.51]	20.23 [18.04,22.42]	20.45 [19.02,21.88]	43.94 [42.22,45.66]	42.87 [40.04,45.70]	43.01 [41.14,44.88]	72.29 [71.15,73.43]	74.23 [73.19,75.27]	73.21 [72.08,74.34]	3.18	3.22	3.20
Qwen3.5-27B	23.87 [22.23,25.51]	23.45 [21.90,25.00]	23.71 [22.11,25.31]	48.62 [44.50,52.74]	47.66 [45.78,49.54]	47.77 [45.89,49.65]	70.29 [69.15,71.43]	71.55 [70.50,72.60]	70.66 [69.53,71.79]	3.08	3.12	3.10
Gemma-3-12B-it	63.89 [49.44,78.34]	70.76 [48.44,93.08]	52.35 [47.16,57.54]	81.90 [70.19,93.61]	89.24 [74.14,104.34]	74.34 [71.41,77.27]	50.47 [49.32,51.62]	52.39 [51.28,53.50]	50.84 [49.66,52.02]	2.00	2.03	2.04
Gemma-3-4B-it	134.82 [105.35,164.29]	119.64 [75.42,163.86]	72.38 [66.21,78.55]	110.31 [95.60,125.02]	121.18 [95.92,146.44]	94.34 [87.92,100.76]	39.96 [38.90,41.02]	42.50 [41.48,43.52]	40.34 [39.33,41.35]	1.48	1.55	1.62
InternVL3.5-8B	116.37 [103.99,128.75]	116.87 [105.2,128.5]	104.01 [93.5,114.5]	127.46 [108.88,146.04]	131.70 [113.5,136.70]	118.61 [112.98,124.24]	12.97 [12.61,13.33]	17.30 [16.12,18.49]	11.71 [11.35,12.07]	1.01	1.01	1.00
InternVL3.5-4B	118.44 [111.20,125.68]	156.92 [148.19,165.65]	115.40 [105.0,125.8]	134.02 [125.27,142.77]	172.92 [163.73,182.11]	137.74 [133.19,142.29]	17.32 [17.00,17.64]	16.43 [16.12,16.74]	7.27 [7.04,7.50]	1.02	1.00	1.00
InternVL3.5-2B	132.73 [125.69,139.77]	176.70 [165.69,187.71]	136.05 [128.00,144.10]	153.86 [146.29,161.43]	202.01 [191.93,212.09]	151.79 [144.50,159.08]	16.36 [16.04,16.68]	15.57 [15.25,15.89]	11.69 [11.35,12.03]	1.01	1.00	1.00
<i>LoRA Fine-Tuning (P1 Prompt)</i>												
Qwen3.5-9B (FT)		21.53 [19.82,23.24]			40.16 [38.41,41.91]			75.93 [74.98,76.88]			3.25	
Gemma-3-12B-it (FT)		28.91 [25.84,31.98]			44.29 [40.88,47.70]			72.79 [71.44,74.14]			2.85	
InternVL3.5-8B (FT)		45.81 [41.37,50.25]			74.60 [69.28,79.92]			59.07 [57.24,60.90]			1.85	
Gemma-3-4B-it (FT)		50.72 [45.11,56.33]			65.82 [60.14,71.50]			69.50 [67.88,71.12]			1.70	
InternVL3.5-4B (FT)		51.73 [46.82,56.64]			77.42 [71.33,83.51]			51.53 [49.61,53.45]			1.50	
InternVL3.5-2B (FT)		51.96 [47.88,56.04]			82.51 [76.02,89.00]			52.99 [51.04,54.94]			1.45	
<i>Conventional OCR Baselines (non-generative, prompt-independent)</i>												
EasyOCR		48.43			89.26			48.45			1.98	
Tesseract		98.46			110.05			3.88			1.01	
Surya		191.32			160.48			11.52			1.02	

Table 1: BANGLAWILD leaderboard. Zero-shot VLMs are evaluated under three prompts on $n = 1,014$ images. **Bold** marks the best of the two *deployment-realistic* prompts (P1, P2) per (model, metric); ties are bolded jointly. †P3 is an *oracle* condition that prepends gold CONTEXT/CONTENT metadata unavailable at inference time and reported for reference only. Subscripts in brackets denote 95% bootstrapped confidence intervals over the test set (1,000 resamples). CER and WER may exceed 100% when models emit long off-target outputs. Judge scores are on a 1–5 scale from an LLM-as-a-Judge protocol (Appendix F); the primary judge is Gemini 2.5 Pro for all models, while Gemini entries marked with * are re-scored by Claude Sonnet 4.6 as an out-of-family judge to control for in-family bias. CER is computed over Unicode code points after NFC normalization.

and Gemini 2.5 Pro from $\{3.68, 3.72, 3.75\}$ to $\{3.58, 3.67, 3.63\}$, implying in-family inflation of 0.16–0.17 points for Flash and 0.05–0.12 for Pro on the 1–5 scale. We therefore report the Sonnet-judged scores for both Gemini models in Tab. 1; the remaining thirteen models retain Gemini Pro scores because full cross-family re-scoring was beyond our budget. This makes the CER-judge inversion at the top conservative: correcting Sonnet 4.6 for its own potential family bias would require an out-of-family Sonnet judge, so its reported semantic advantage is more likely understated than overstated. More generally, judge-based comparisons between Gemini and non-Gemini models should be interpreted with possible residual in-family bias, and we treat judge scores as complementary to CER rather than a standalone ranking metric.

Where the errors live. Tabs. 11, 12, and 14 characterize the failure surface. *Marginals* (Tab. 11): on the Context axis (Gemini 2.5 Flash, P1), CER increases from PRINTED MATERIAL (7.70) to PACKAGING (10.26), SIGNBOARD (13.69), WALL (13.74), and VE-

HICLE (30.35), a $3.9\times$ span separating planar from hand-painted curved surfaces. On the Content axis, CER rises from INSTRUCTIONAL (8.76) to RELIGIOUS (11.94), ADVERTISEMENT (12.42), CREATIVE (13.39), POLITICAL (17.42), and COLLOQUIAL (21.69), a $2.5\times$ span reflecting increasing typographic variability. Both rankings are consistent across Claude Sonnet 4.6 and Qwen3.5-9B, indicating dataset rather than model effects. *Joint cells* (Tab. 14): the marginals compound. The easiest cells are PRINTED MATERIAL×INSTRUCTIONAL (CER 2.33) and SIGNBOARD×INSTRUCTIONAL (5.81); the hardest are VEHICLE×CREATIVE (48.22–60.38), VEHICLE×COLLOQUIAL (33.03–47.48), and WALL×COLLOQUIAL (30.05–34.91). The effects are superadditive: VEHICLE×CREATIVE (60.38 at P1) exceeds the sum-minus-mean prediction from the marginals, showing that difficulty arises from the interaction of hand-painted surfaces and informal language rather than either factor alone. *Diagnostic axes* (Tab. 12): CURVED TEXT increases CER by $2.2\times$ for both Gem-

ini 2.5 Flash (13.07 $\rightarrow$ 28.69) and Qwen3.5-9B (18.88 $\rightarrow$ 41.98); HEAVY OCCLUSION by 1.6 $\times$ and 1.3 $\times$; and CLUTTERED BACKGROUND by 1.8 $\times$ and 1.4 $\times$, respectively. In contrast, ARTISTIC versus PRINTED fonts increase CER by only 1.4 $\times$ and 1.2 $\times$. The dominant challenges are layout non-linearity and communicative informality, rather than scene text augmentation targets such as perspective, occlusion, or decorative typography.

6 Error Analysis

Aggregate CER/WER quantify *how far* predictions deviate from ground truth, but not *how* they fail. We therefore classify every error into fifteen mutually exclusive categories (Tab. 2) using a deterministic classifier applied to four VLM families (closed-source Gemini-2.5-Flash and Claude Sonnet 4.6, open-source Qwen3.5-9B and Gemma-3-12B-it) and three OCR baselines (Tesseract, EasyOCR, Surya). Unless otherwise noted, the analysis focuses on Gemini-2.5-Flash and Qwen3.5-9B as representative strong systems; the full per-model breakdown is given in Tab. 2.

Bengali scene text failure is dominated by visual mis-recognition, not orthographic confusion. Lexical substitution, single-character visual misread, and multi-character word corruption, all reflecting *vision-encoder* failures, account for $\sim 56\%$ of Gemini-2.5-Flash errors and $\sim 60\%$ of Qwen3.5-9B errors across all prompts. In contrast, the *orthographic* categories emphasized in prior Indic OCR work (matra substitution, consonant homophones, conjunct/hasanta errors, nasal diacritics, and compound orthographic edits) contribute only $\sim 13\%$ and $\sim 10\%$, yielding visual: orthographic ratios of 4.4 $\times$ and 5.8 $\times$, respectively. This reverses the prevailing assumption in Bengali OCR: for generative VLMs, the bottleneck is no longer decoding complex orthography but *recognizing glyphs* on cluttered, hand-painted, or oblique surfaces. The qualitative errors are consistent with visual rather than linguistic failures: ফ্লাটটি (*phlyāṭṭi*) $\rightarrow$ ফ্যাক্টরি (*phyākṭari*) and কপাট (*kapāṭa*) $\rightarrow$ লোহ (*loha*) are neither homophones, near-conjuncts, nor plausible language-prior substitutions, but visual misreadings. This pattern is architecture-independent: on EasyOCR, the only conventional baseline that reliably reads Bengali, visual errors outnumber orthographic ones by 3.7 $\times$ (Tab. 2), matching the VLMs. The bottleneck therefore lies in the in-the-wild imagery rather than

any specific model class.

The classical conjunct bottleneck has largely closed. Conjunct/hasanta errors account for only 1.0–1.6% of errors, despite CRNN-era Bengali OCR treating *yuktakshara* recognition as the primary structural challenge and prior Indic benchmarks (Nayef et al., 2019; Gunna et al., 2022) reporting only aggregate metrics that do not isolate conjunct errors. Modern VLMs appear to have learned Bengali conjunct complexity through large-scale pre-training. Even on EasyOCR, conjunct/hasanta errors are more frequent (4.63% of residual errors, a ~ 3 –4 $\times$ higher share than VLMs) but remain secondary to segmentation and lexical substitution. The bottleneck has therefore diminished across model classes, not just VLMs.

Cross-script contamination is the only failure mode affected by prompting. Among the fifteen error categories, only script contamination changes measurably with prompt language: it drops from 6.8% $\rightarrow$ 5.5% on Gemini and 5.6% $\rightarrow$ 4.8% on Qwen under P2, then rebounds at P3 when English is reintroduced. Lexical substitution, grapheme misread, matra error, and conjunct error remain stable within $\pm 1.5\text{pp}$ across P1/P2/P3 for both models. This suggests prompt language gates the *output language head* (Bengali conditioning reduces probability mass on residual Devanagari or Latin transliteration pathways) but not the visual encoder, where the dominant errors occur. This supports the visual-bottleneck claim above and bounds what prompt engineering alone can achieve on this benchmark.

Tokenization and Unicode artefacts inflate WER without reducing readability, and vary by model. Word-boundary segmentation (12–19% of errors) and Unicode/punctuation normalization (5–11%) together account for 22–25% of all errors while leaving readable content unchanged: an attached এনা (*enā*) $\rightarrow$ এন (*ena*) এ (*e*), a typographic ” vs. “, or an NFC-equivalent code point. These categories explain much of the Gemini-Qwen gap in aggregate metrics: Qwen3.5-9B produces 2.3 $\times$ more Unicode-normalization noise than Gemini-2.5-Flash ($\sim 11\%$ vs. $\sim 5\%$), despite visually identical output. They also explain the Sonnet-Flash judge inversion in §5: Sonnet’s higher judge score is consistent with more spelling-normalized yet meaning-preserving transcriptions, which edit-distance metrics penalize but the judge does not. WER-based deployment decisions for

Error type	Conventional OCR			Gemini-2.5-Flash			Qwen3.5-9B			Claude Sonnet 4.6			Gemma-3-12B-it		
	Tesseract	EasyOCR	Surya	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
Lexical word substitution	11.10	26.56	16.09	26.9	27.5	27.2	24.8	24.1	24.3	21.95	23.59	23.35	43.52	41.11	43.14
Visual grapheme misread	2.37	10.11	4.88	18.9	17.3	18.7	22.3	22.8	23.0	17.50	17.35	17.43	13.71	11.94	13.79
Word-boundary segmentation	17.43	27.78	11.33	17.3	19.3	16.5	11.7	13.0	12.2	14.45	15.89	13.93	13.52	18.32	13.39
Multi-character word corruption	1.67	10.34	2.93	10.4	10.2	10.3	12.6	12.5	12.5	9.41	9.21	9.23	11.08	10.77	11.40
Vowel-sign (matra) substitution	0.23	4.25	0.30	7.5	6.8	7.2	7.2	7.4	7.1	6.41	6.02	6.01	4.19	3.76	4.47
Script contamination	1.50	1.01	51.83	6.8	5.5	6.7	5.6	4.8	5.4	8.23	6.18	7.27	8.57	10.17	8.38
Unicode normalization	0.40	2.01	1.83	4.8	5.3	5.6	11.0	10.7	11.0	16.58	16.26	16.63	1.53	0.95	1.51
Consonant homophone	0.03	1.16	0.12	2.2	2.4	2.2	1.0	0.9	1.0	1.20	1.38	1.48	0.34	0.36	0.29
Conjunct / hasanta error	0.06	4.63	0.06	1.4	1.4	1.6	1.0	1.1	1.0	1.25	1.14	1.28	0.26	0.24	0.30
Compound orthographic	0.06	1.34	0.00	1.0	1.1	0.9	0.9	0.8	0.9	0.76	0.77	0.82	0.69	0.63	0.78
Semantic hallucination	2.32	4.38	2.62	0.6	0.9	0.8	1.0	1.0	0.7	1.23	1.01	1.31	2.06	1.44	1.96
Numeric / temporal error	0.03	1.03	0.00	1.2	1.2	1.3	0.3	0.4	0.3	0.27	0.34	0.33	0.14	0.15	0.11
Nasal diacritic error	0.00	1.20	0.00	0.7	0.6	0.7	0.3	0.4	0.3	0.71	0.72	0.76	0.11	0.10	0.18
Truncation / no output	62.80	3.16	7.95	0.1	0.1	0.1	0.1	0.1	0.2	0.03	0.05	0.05	0.26	0.06	0.29
Spurious extra content	0.03	1.04	0.06	0.1	0.2	0.1	0.1	0.1	0.1	0.03	0.08	0.11	0.02	0.02	0.02
Exact-match accuracy (%)	0.16	4.69	0.63	35.6	35.9	35.3	16.1	16.3	16.1	25.81	26.09	26.60	4.42	4.54	5.41

Table 2: Unified error taxonomy on BANGLAWILD for three conventional OCR baselines and four VLM families across the three prompts. OCR values are % of residual errors and are prompt-independent. VLM values are % of errors per prompt. A detailed explanation of the error classification protocol can be found in Appendix H.

Bengali scene text should therefore account for each model’s share of cosmetic errors.

Two small categories carry disproportionate deployment risk. Semantic hallucination (0.6-1.0%) and numeric/temporal substitution (0.3-1.3%) are individually rare, but uniquely produce fluent, plausible Bengali that cannot be recognized as incorrect from the output alone. Hallucinations preserve discourse type while replacing lexical identity: ছাত্র ইউনিয়ন (*chātra iuniyana*) → শুভ জন্মদিন (*śubha janmadina*), হর্ণ (*harṇa*) → শুভ নববর্ষ (*śubha nababarṣa*). Numeric substitutions silently rewrite prices, years, and phone numbers (১৪৩৩ (1433) → ১৪৩০ (1430), ২০২৫ (2025) → ২০২৩ (2023)) and are $\sim 4\times$ more common in Gemini-2.5-Flash than Qwen3.5-9B (1.2% vs. 0.3%), with rates unchanged across prompts, suggesting a training-time prior over plausible Bengali calendar years rather than a prompt effect. Although neither category dominates CER, they matter most for verbatim applications such as transcribing addresses, receipts, election material, and medical packaging. They also explain cases where CER is low but LLM-judge scores are poor: the judge detects semantic mismatches against the reference, whereas users see only fluent but incorrect text. Additional labelled examples are provided in Appendix I.

Per-word comparisons reveal shared orthographic confusions and model-specific hallucinations. Tab. 13 compares ground-truth Bengali words, their corrected forms, and the outputs of four leading systems on identical inputs. Two patterns emerge: 1. Most disagreements reflect orthographic contrasts subtle to non-Bengali readers: retroflex versus dental nasal (ণ/ন; *n̥/n*), long versus short vowel signs (ী/ি; *ī/i*), retroflex ver-

sus dental sibilant (ষ/স; *ṣ/s*), and aspirated versus unaspirated retroflex (ঢ়/ড়; *ḍh/r̥*). Although CER penalizes these equally, models diverge: for সুবিদা (*subidā*), Qwen3.5-35B-A3B alone matches the ground truth, Qwen3.5-9B and Claude Sonnet 4.6 produce the corrected সুবিভা (*subibhā*), and Gemini 2.5 Flash outputs a third variant, সুবিধা (*subidhā*). 2. Hallucinations are model-specific: for হর্ণ (*harṇ*), Qwen3.5-9B outputs *jana*, Qwen3.5-35B-A3B outputs *tarka*, Gemini 2.5 Flash hallucinates শুভ নববর্ষ (*śubha nababarṣa*), and Claude Sonnet 4.6 produces the corrected হর্ণ (*harn*), a text-book correction error revealed by the dual annotation in §3. The diversity of outputs on identical inputs suggests distinct language-prior pathways rather than shared visual confusion.

7 Conclusion

We introduced BANGLAWILD, the first in-the-wild Bengali scene text recognition benchmark to pair dual verbatim/standard annotation with evaluation of both conventional OCR and generative VLMs on the same data. Across 15 VLMs, 3 conventional OCR systems, 3 prompting strategies, 6 LoRA-adapted variants, and a 15-class error taxonomy, our key finding inverts a long-standing CRNN-era assumption: visual mis-recognition, not orthographic confusion, dominates Bengali scene text recognition failure, a pattern that holds across VLMs and a conventional OCR baseline, and the classical conjunct bottleneck has largely closed. The dual annotation additionally surfaces generative-model-specific failures, silent over-correction and fluent hallucination, that conventional OCR structurally cannot exhibit. We hope our dataset, taxonomy, and findings will foster future research toward more visually robust Indic-script recognition systems.

Limitations

Despite the contributions of this work, several limitations remain. The benchmark covers Bengali text only; code-mixed Bengali–English images were excluded at the filtering stage to keep the evaluation target unambiguous, leaving the code-mixing problem to future purpose-built resources. The categorical distribution reflects natural abundance rather than balance: WALL dominates CONTEXT and six of the thirty possible (CONTEXT, CONTENT) combinations contain no images, so per-cell results in the sparser categories carry higher variance than the macro-averages suggest. Our LoRA experiments adapt six open-source models on $\sim 1,268$ training images and use a single random seed under our compute budget; the negative result on already-competent models tests targeted small-scale adaptation, not whether full-scale pre-training revision would help. The LLM-as-a-Judge is itself a frontier VLM and may share blind spots with the systems it scores, particularly leniency toward fluent-but-incorrect Bengali; we validate against native-annotator re-scoring ($\rho = 0.89$). To mitigate in-family bias (Panickssery et al., 2024; Li et al., 2025), we re-scored both Gemini-family systems with Claude Sonnet 4.6 as an out-of-family judge and report those values in Tab. 1; budget constraints prevented out-of-family re-scoring of the remaining thirteen models, so residual in-family inflation cannot be excluded for any non-Gemini cell. We report judge scores alongside, not in place of, edit-distance metrics. Finally, our fifteen-class error taxonomy was applied to four VLM families (Gemini 2.5 Flash, Qwen3.5-9B, Claude Sonnet 4.6, and Gemma-3-12B-it) and three conventional OCR baselines (Tesseract, EasyOCR, Surya) using an identical deterministic classifier (Appendix H); the detailed per-category analysis in §6 focuses on Gemini 2.5 Flash and Qwen3.5-9B as representative strong systems, and the reported percentages should not be assumed to transfer without re-classification to systems with substantively different failure profiles.

Ethical Statement

This work involves the creation and use of a Bengali scene text benchmark sourced from public-facing physical surfaces. Trained collaborators captured all images in public space consistent with local norms for street photography; the benchmark targets text, not people, and where human faces or

vehicle registration plates appeared incidentally in the frame, we blurred them and verified that the remaining content is non-uniquely-identifying prior to release. Vulgar, offensive, and entirely unreadable content was excluded at the filtering stage. We retain politically sensitive material such as protest slogans, election posters, and religious verses, because excluding it would systematically misrepresent the visual environment the benchmark is intended to characterize; such content is transcribed factually and without endorsement. Annotators from Bangladesh (three native Bengali-speaking transcribers, one expert verifier, and two additional annotators for judge validation) were compensated at standard per-sample rates and consented to the release of their adjudicated annotations. We acknowledge that VLMs evaluated on this benchmark may produce inaccurate, biased, or culturally inappropriate outputs; the benchmark exists to surface such failures, not to validate deployment. The failure modes we document carry concrete downstream risks if such systems are deployed for verbatim Bengali transcription: misread addresses can cause misdelivery, silently altered numerals can corrupt prices, dates, and dosages on medical packaging, and hallucinated or over-corrected text on political and religious surfaces can misattribute or distort sensitive content. These risks fall disproportionately on a low-resource-language population with limited recourse, which is precisely why we release BANGLAWILD as a held-out diagnostic resource rather than a training corpus; the release license (CC BY-NC 4.0) reflects this intended use.

References

- American Library Association and Library of Congress. 2017. *Bengali Romanization Table*. ALA-LC Romanization Tables.
- Anthropic. 2025. Claude haiku 4.5 system card. <https://www.anthropic.com/claude-haiku-4-5-system-card>.
- Anthropic. 2026. Claude sonnet 4.6 system card. <https://www.anthropic.com/claude-sonnet-4-6-system-card>.
- Anik De, Abhirama Subramanyam Penamakuri, Rajeev Yadav, Aditya Rathore, Harshiv Shah, Devesh Sharma, Sagar Agarwal, Pravin Kumar, and Anand Mishra. 2025. Bharat scene text: A novel comprehensive dataset and benchmark for indian language scene text understanding. *arXiv preprint arXiv:2511.23071*.

- Ali Faraz, Akash, Shaharukh Khan, Raja Kolla, Akshat Patidar, Suranjan Goswami, Abhinav Ravi, Chandra Khatri, and Shubham Agarwal. 2025. Indicvisionbench: Benchmarking cultural and multilingual understanding in vlms. *arXiv preprint arXiv:2511.04727*.
- Ling Fu, Zhebin Kuang, Jiajun Song, Mingxin Huang, Biao Yang, Yuzhe Li, Linghao Zhu, Qidi Luo, Xinyu Wang, Hao Lu, Zhang Li, Guozhi Tang, Bin Shan, Chunhui Lin, Qi Liu, Binghong Wu, Hao Feng, Hao Liu, Can Huang, and 5 others. 2025. Ocrbench v2: An improved benchmark for evaluating large multimodal models on visual text localization and reasoning.
- Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92.
- Gemini Team, Google, Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, and 1 others. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*.
- Sanjana Gunna, Rohit Saluja, and C.V. Jawahar. 2022. Transfer learning for scene text recognition in Indian languages. *arXiv preprint arXiv:2201.03180*.
- Md. Yearat Hossain and Tanzilur Rahman. 2023. A crowdsourcing based framework for bengali scene text data collection and detection. *Computers and Electrical Engineering*, 109:109025.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *ICLR*.
- Gemma Team Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, Louis Rouillard, Thomas Mesnard, Geoffrey Cideron, Jean-Bastien Grill, Sabela Ramos, Edouard Yvinec, Michelle Casbon, Etienne Pot, Ivo Penchev, Gael Liu, and 191 others. 2025. *Gemma 3 technical report*. *ArXiv*, abs/2503.19786.
- Dawei Li, Renliang Sun, Yue Huang, Ming Zhong, Bohan Jiang, Jiawei Han, Xiangliang Zhang, Wei Wang, and Huan Liu. 2025. Preference leakage: A contamination problem in LLM-as-a-judge. *arXiv preprint arXiv:2502.01534*.
- Yuliang Liu, Zhang Li, Mingxin Huang, Biao Yang, Wenwen Yu, Chunyuan Li, Xu-Cheng Yin, Chenglin Liu, Lianwen Jin, and Xiang Bai. 2024. *Ocrbench: on the hidden mystery of ocr in large multimodal models*. *Science China Information Sciences*, 67(12).
- Meta AI. 2025. The Llama 4 herd: Native multimodality, mixture-of-experts and long context. <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>. Open-weight release of Scout and Maverick variants, April 2025.
- Hasan Murad and Mohammed Eunus Ali. 2023. Towards detecting, recognizing, and parsing the address information from Bangla signboard: A deep learning-based approach. *arXiv preprint arXiv:2311.13222*.
- Nibal Nayef, Yash Patel, Michal Busta, Pinaki Nath Chowdhury, Dimosthenis Karatzas, Wafa Khelif, Jiri Matas, Umapada Pal, Jean-Christophe Burie, Chenglin Liu, and Jean-Marc Ogier. 2019. ICDAR2019 robust reading challenge on multi-lingual scene text detection and recognition — RRC-MLT-2019. In *Proc. of the International Conference on Document Analysis and Recognition (ICDAR)*, pages 1582–1587.
- Surapon Nonesung and 1 others. 2025. ThaiOCR-Bench: A task-diverse benchmark for vision-language understanding in thai. In *Proceedings of AACL*.
- OpenAI. 2024. *Gpt-4o system card*.
- OpenAI. 2026. Introducing GPT-5.4 mini and nano. <https://openai.com/index/introducing-gpt-5-4-mini-and-nano/>.
- Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. Llm evaluators recognize and favor their own generations. *arXiv preprint arXiv:2404.13076*.
- Qwen Team. 2026. *Qwen3.5: Towards native multimodal agents*.
- Md. Ataur Rahman, Nazifa Tabassum, Mitu Paul, Riya Pal, and Mohammad Khairul Islam. 2022. *Bn-htrd: A benchmark dataset for document level off-line bangla handwritten text recognition (htr) and line segmentation*. *Preprint*, arXiv:2206.08977.
- Jin Seong, Wencke Liermann, Minho Kim, Jong hun Shin, and Soojong Lim. 2026. *When vlms 'fix' students: Identifying and penalizing over-correction in the evaluation of multi-line handwritten math ocr*. *Preprint*, arXiv:2604.22774.
- Muhammad Abdullah Sohail, Salaar Masood, and Hamza Iqbal. 2024. Deciphering the underserved: Benchmarking llm ocr for low-resource scripts. *arXiv preprint arXiv:2412.16119*.
- Evangelia Spiliopoulou and 1 others. 2025. Play favorites: A statistical method to measure self-bias in LLM-as-a-judge. *arXiv preprint arXiv:2508.06709*.
- Jingqun Tang, Qi Liu, Yongjie Ye, Jinghui Lu, Shu Wei, Chunhui Lin, Wanqing Li, and 1 others. 2024. MTVQA: Benchmarking multilingual text-centric visual question answering. *arXiv preprint arXiv:2405.11985*.

Leandro von Werra, Younes Belkada, Lewis Tunstall, Edward Beeching, Tristan Thrush, Nathan Lambert, Shengyi Huang, Kashif Rasul, and Quentin Galouédec. 2020. [TRL: Transformers Reinforcement Learning](#).

Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, and 1 others. 2025. [InternVL3.5: Advancing open-source multimodal models in versatility, reasoning, and efficiency](#). *arXiv preprint arXiv:2508.18265*.

Koki Wataoka, Tsubasa Takahashi, and Ryokan Ri. 2024. Self-preference bias in LLM-as-a-judge. *arXiv preprint arXiv:2410.21819*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhonghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

A Category Definitions

While the main text gives informal descriptions of each category for the reader, this section formally defines each value of the `CONTEXT` and `CONTENT` axes to support reproducibility and annotation extension.

A.1 Context (5 values)

- **Vehicle.** Text painted, decaled, or otherwise affixed to buses, trucks, three-wheelers, and similar conveyances. Typically rendered in stylised hand-painted vehicle lettering with strong artistic flourishes.
- **Signboard.** Shop and storefront signage, including hand-painted, vinyl-printed, and back-lit varieties. Text is the dominant visual element of the surface.
- **Packaging.** Consumer product packaging, predominantly processed food and household goods. Text is typically small, dense, and embedded in busy commercial designs.
- **Wall.** Text appearing on built surfaces such as painted notices, posters affixed to walls, stencilled markings, and free-painted wall-text. The surface itself is not portable.
- **Printed Material.** Banners, leaflets, and posters that are free-standing, strung, or hand-held, rather than affixed to a wall surface.

A.2 Content (6 values)

- **Religious.** Verses, invocations, names, and aphorisms drawn from religious traditions. `BANGLAWILD` includes content from Islamic, Hindu, Buddhist, and Christian traditions.
- **Political.** Slogans, election material, and protest-related text. Identification is based on overtly political content rather than on the surface or the speaker.
- **Advertisement.** Commercial promotions for products and services, including food, construction materials, telecommunications, and consumer goods.
- **Instructional.** Directive or informational notices in public space: prohibitions, warnings, requests for assistance, and traffic/safety instructions.
- **Creative.** Song lyrics, poetry excerpts, literary quotations, and aphorisms appearing in public visual culture (rather than in printed books).
- **Colloquial.** Everyday, miscellaneous public-space text not fitting the other categories, such as names, greetings, and idiosyncratic markings. Defined explicitly as a residual category so that ambiguous text is not forced into ill-fitting labels.

The six `CONTENT` categories are mutually exclusive. Where an image straddles two values (e.g., a religious aphorism deployed as a political slogan), annotators were instructed to select the value that better captures the speaker’s primary communicative intent and to flag the image for adjudication.

B Annotation Rubric

We reproduce the rubric given to annotators. The rubric was developed iteratively over a pilot annotation of approximately 200 images before being frozen for the main annotation pass.

Transcription.

1. Transcribe verbatim, not normatively. Type exactly what is written in the image, including spelling errors, unconventional spacing, or non-standard vowel sign placement. Do not “fix” misspellings, even obvious ones.
2. If the in-image text is misspelled, additionally fill the `standard_form` field with the ortho-

Diagnostic attribute	Count
CURVED TEXT - No	2335
CURVED TEXT - Yes	200
FONT STYLE - Handwritten	1436
FONT STYLE - Printed	644
FONT STYLE - Artistic	455
OCCCLUSION - None	2009
OCCCLUSION - Partial	503
OCCCLUSION - Heavy	23
BACKGROUND COMPLEXITY - Clean	2158
BACKGROUND COMPLEXITY - Moderate	279
BACKGROUND COMPLEXITY - Cluttered	98

Table 3: Distribution of diagnostic attributes in BANGLAWILD.

graphically standard form. Leave it blank if no deviation is present.

3. Transcribe only Bengali text. If the image also contains English, numerals, or other-script content, transcribe only the Bengali. Flag the image for the code-mixed exclusion filter if Bengali is not the dominant content.
4. For multi-line text, transcribe top-to-bottom, left-to-right within each line. For text on curved baselines, follow the baseline in the natural reading direction.
5. Preserve punctuation as it appears. Use the Bengali danda where present rather than a Latin period.
6. If uncertain about a character, mark it with [?]; the expert verifier will adjudicate. Do not guess. If the text is entirely unreadable, flag the image for the unreadable-text exclusion filter.

Categorical labels. CONTEXT answers “what physical medium does this text appear on?” and is determined by the surface. CONTENT answers “what is the text trying to communicate?” and is determined by intent independent of the surface.

Diagnostic attributes. Beyond the categorical axes, each image carries four diagnostic labels targeting known failure modes in scene text recognition: *Curved Text*: label *Yes* iff the text layout itself is non-linear in the original physical surface; a straight-line sign photographed at an oblique angle remains *No*. *Font Style*: *Printed* (machine-generated, uniform stroke), *Handwritten* (naturalistic stroke variation), *Artistic* (stylised, decorative, readability traded for aesthetics). *Occlusion*:

Reliability metric	Score
Mean pairwise CER (transcription)	3.2%
Cohen’s κ CONTEXT	0.93
Cohen’s κ CONTENT	0.88
Cohen’s κ CURVED TEXT	0.89
Cohen’s κ FONT STYLE	0.82
Cohen’s κ OCCCLUSION	0.84
Cohen’s κ BACKGROUND COMPLEXITY	0.81

Table 4: Inter-annotator agreement on BANGLAWILD. Conventionally, $\kappa, \alpha \geq 0.80$ indicates strong agreement.

None, *Partial* (obscured but inferable from context), *Heavy* (recovery difficult even for a native reader). *Background Complexity*: *Clean* (uniform), *Moderate* (patterned, with text-background separability preserved), *Cluttered* (multiple textures or overlapping objects competing with the text). The complete distribution of the diagnostic attributes can be found in Tab. 3

C Inter-Annotator Agreement

Tab. 4 reports the full reliability breakdown referenced in § 3.

D Datasheet

We follow the datasheet template of Gebru et al. (2021); we summarise the essential items here.

Motivation. BANGLAWILD was created to enable rigorous evaluation of vision-language models on Bengali scene text under deployment-realistic conditions, since existing Bengali OCR resources are dominated by cropped, rectified, or born-digital text. The benchmark is intended as a held-out evaluation resource, not a training corpus. Affiliation and funding details are withheld during double-blind review.

Composition. 2,535 image instances, each a JPEG paired with a verbatim gold transcription, CONTEXT and CONTENT labels, four diagnostic attribute labels, and, where applicable, an orthographically standard form. The dataset is a directed (non-probabilistic) sample, drawn from a candidate pool of 3,210 and filtered as described in § 3.

Collection. Each image was captured directly by a trained collaborator using a consumer mobile-phone camera in the physical setting where the text appeared. Collection spanned different regions of Bangladesh to capture variety in signage conventions, typographic styles and surface types, ensuring that the captured surfaces reflect naturally occurring Bengali scene text rather than the visual

environment of any single locality. Camera models and resolutions were not standardised by design. The data captures public surfaces, not people; where individuals appeared incidentally in the frame, faces were blurred and verified non-uniquely-identifying.

Preprocessing. Quality-driven filtering and PII treatment of incidentally visible faces and license plates. No image enhancement, cropping, rectification, or denoising was applied to the released images.

Uses. Evaluation of Bengali scene text detection, recognition, end-to-end OCR, and VLM visual reasoning over text-bearing images. The diagnostic attributes additionally support targeted robustness studies (e.g., recognition under occlusion, on curved layouts, or against cluttered backgrounds). The dataset should not be used to train commercial OCR or VLM systems.

Sensitive content. Vulgar and offensive content was excluded at the filtering stage. The POLITICAL category contains slogans some readers may find ideologically charged; we retain such content because excluding it would systematically misrepresent the visual environment the benchmark is intended to characterize. No content in the released benchmark depicts violence, hate speech, or targeted harassment.

Distribution & maintenance. The dataset will be released under the Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license for non-commercial research use. The current release is **v1.0**; errata will be issued as point releases preserving backward compatibility of the `file_name` and `Serial` fields. Hosting, maintenance, and contribution channels will be announced in the camera-ready version.

E Prompts

Prompt	Description
English baseline (P1)	Transcribe-only instruction issued in English. Image-only; deployment-realistic.
Bengali baseline (P2)	The P1 instruction translated into Bengali script. Image-only; deployment-realistic.
Structured metadata (P3)	Gold <code>CONTEXT</code> and <code>CONTENT</code> labels prepended to a revised transcription instruction. Oracle condition: the metadata is annotator-provided and unavailable at inference.

Table 5: Prompting strategies used in the zero-shot benchmark. P3 reuses the P1 instruction verbatim and only varies the preceding context. P1/P2 are image-only and form the deployment-realistic setting; P3 is an oracle condition reported for reference and excluded from headline and best-prompt comparisons.

P1: Base/Zero-Shot Prompt

Read and transcribe exactly the text in this image.
Output only the text, with no additional commentary, explanation, or formatting.

P2: Bengali-Translated Base/Zero-Shot Prompt

এই ছবিতে যে লেখা আছে তা হুবহু পড়ুন এবং প্রতিলিপি করুন। (*ei chabite ye lekhā āche tā hubahu paṛuna ebaṃ pratilipi karuna.*) শুধুমাত্র লেখাটি লিখুন, কোনো অতিরিক্ত মন্তব্য, ব্যাখ্যা বা ফরম্যাটিং ছাড়া। (*śudhumātra lekhāṭi likhuna, kono atiri ta mantabya, byākhyā bā pharamyāṭim chārā.*)

P3: Structured Metadata-Augmented Prompt

This image is annotated with structured metadata:
Context: {context}
Content: {content}
Context refers to the physical location or medium where the text appears. *Content* refers to the communicative intent of the text.
Based on this information, read and transcribe exactly the text in this image.
Output only the text, with no additional commentary, explanation, or formatting.

LLM-as-a-Judge Prompt

You are an expert Bengali (Bengali) evaluator judging transcriptions of stylized in-the-wild Bengali text.

You will see:

- **GROUND_TRUTH** — human-annotated transcription (authoritative)
- **MODEL_OUTPUT** — system’s transcription attempt

Judge how well **MODEL_OUTPUT** matches **GROUND_TRUTH**. Accept valid Bengali spelling variants and equivalent Unicode encodings. Ignore whitespace differences.

Score on a 1–5 integer scale:

- 5 = perfect / fully equivalent
- 4 = minor errors, meaning preserved
- 3 = several errors, gist recoverable
- 2 = major errors, only fragments usable
- 1 = wrong, empty, or hallucinated

Output ONLY a single digit (1, 2, 3, 4, or 5). No other text.

GROUND_TRUTH: {ground_truth}

MODEL_OUTPUT: {model_output}

F LLM-as-a-Judge Protocol

Motivation. CER, WER, and 1–NED are surface-level edit-distance metrics: they penalize every character deviation uniformly, regardless of whether the deviation preserves meaning. In Bengali scene text, this is a particularly acute limitation. A model that emits *sbādhina* (স্বাধীন) for the ground truth *sbādhina* (স্বাধীন) differs in only the long/short *i* vowel sign, yet both forms denote the same lexeme and a native reader treats them as equivalent. Conversely, a low-CER output that silently substitutes a confusable consonant (ণ → ন) can shift the lexical identity entirely. To complement the edit-distance metrics with a meaning-aware signal, we additionally score every (ground truth, model output) pair with an LLM-as-a-Judge protocol.

Judge model and decoding. We use Gemini 2.5 Pro as the primary judge for all fifteen systems across all three prompts, queried via the Google API with temperature = 0 and a max_output_tokens cap of 4. To control for in-family bias on the two Gemini systems being scored, we additionally re-score Gemini 2.5

Flash and Gemini 2.5 Pro with Claude Sonnet 4.6 (Anthropic API, identical decoding settings and rubric); the Sonnet judged scores replace the Gemini Pro judged ones for these two systems in Tab. 1. Cross-family re-scoring of the remaining thirteen models was infeasible under our compute and API budget (see Limitations). The judge sees only the ground-truth transcription and the candidate model output; it does not see the source image, the prompting strategy used, or the identity of the model being scored. This isolates judgments to textual equivalence and prevents any visual-grounding bias from leaking into the score.

Rubric. The judge assigns an integer score on a 1–5 Likert scale: 5 denotes fully equivalent transcriptions (modulo valid Bengali orthographic variants and Unicode-equivalent encodings), 4 denotes minor errors that preserve meaning, 3 denotes several errors with the gist still recoverable, 2 denotes major errors with only fragments usable, and 1 denotes wrong, empty, or hallucinated outputs. The judge is instructed to ignore whitespace differences and to accept canonical spelling variants, which makes the score robust to the benign normalization differences that inflate raw CER.

Coverage and aggregation. We score the full $n = 1,014$ test set for every (model, prompt) cell reported in Table 1, yielding $1,014 \times 3$ judgments per zero-shot model and 1,014 judgments per fine-tuned model. The reported **Judge** score is the arithmetic mean of the per-image scores within each cell. Outputs that returned a non-digit response from the judge ($< 0.1\%$ of calls across all models) were re-queried once; any remaining failures were excluded from the mean for that cell.

Validation. To verify that the judge tracks human judgment rather than amplifying a single model’s biases, we sampled 200 (ground truth, model output) pairs stratified across score bands and had two of the native-Bengali annotators who didn’t participate in the annotation process, re-score them blind under the same rubric. They were compensated at standard per-sample rates. The judge agreed with the annotator-majority score within ± 1 on 94.5% of pairs and exactly on 71.0%, with a Spearman rank correlation of $\rho = 0.89$ between judge and annotator-majority scores. This is consistent with the agreement levels reported for LLM-as-a-Judge on text-generation tasks and supports the use of the score as a complementary met-

ric, though not as a replacement for CER/WER on this benchmark.

Limitations. The judge is itself a frontier VLM and may share blind spots with the systems it scores; in particular, it may be lenient toward outputs that drift into plausible but incorrect Bengali, since fluency is a strong prior for any large language model. A second, distinct concern is in-family bias: prior work has shown that LLM judges systematically favor outputs from their own family (Panickssery et al., 2024; Wataoka et al., 2024; Spiliopoulou et al., 2025; Li et al., 2025). We address this for the two Gemini-family systems by re-scoring them with Claude Sonnet 4.6 as an out-of-family judge and using those scores in the main leaderboard, but we were unable to apply out-of-family re-scoring to the remaining thirteen models under our budget, so residual in-family inflation cannot be excluded for any non-Gemini cell. We therefore report judge scores alongside, not in place of, the edit-distance metrics, and treat large disagreements between the two (e.g., a model with moderate CER but a low judge score) as a diagnostic signal of semantic hallucination rather than as a ranking authority.

G Experimental Setup

G.1 Evaluated Models

Our evaluation covers fifteen VLMs across closed-source and open-source families (Tab. 7), together with three conventional OCR baselines (EasyOCR, Tesseract, Surya). The fifteen VLMs are run across all three prompting conditions. Closed-source models are queried via their respective provider APIs with temperature = 0 and a maximum of 150 completion tokens; images are submitted as base64-encoded data URLs. Open-source models are served locally in bfloat16 with greedy decoding and the same 150-token cap. The fine-tuned and zero-shot evaluations share the same held-out test set.

G.2 Fine-Tuning Details

We fine-tune the six open-source models listed in Tab. 1: InternVL3.5-2B, InternVL3.5-4B, InternVL3.5-8B, Gemma-3-4B-it, Gemma-3-12B-it, and Qwen3.5-9B. Closed-source models are excluded since their weights are not publicly accessible. LoRA adapters are attached to the attention projections (q_proj, k_proj, v_proj, o_proj) and MLP projections (gate_proj, up_proj,

down_proj) of the *language-model component only*; the vision encoder and any vision-language adapter remain frozen. Base weights are loaded in bfloat16 without quantization. We use the SFTTrainer from trl (von Werra et al., 2020) with a custom collator that masks the user-prompt tokens so loss is computed only over assistant tokens.

We use a 50/10/40 train/validation/test split, with the same test split used for the zero-shot evaluation. Training runs for up to 10 epochs with early stopping on validation loss (patience = 2). The effective batch size is 16 (per-device 2, gradient accumulation 8), with AdamW, a peak learning rate of 2×10^{-4} , cosine decay, and 10 warmup steps. All runs use a single seed (seed = 3407) on one NVIDIA A100 GPU. Full hyperparameters are listed in Tab. 6.

Hyperparameter	Value
LoRA rank r	16
LoRA alpha α	32
LoRA dropout	0.05
LoRA target modules	$q, k, v, o; up, down, gate$
Max epochs	10
Early stopping patience	2
Batch size (per device)	2
Gradient accumulation	8 (eff. batch = 16)
Optimizer	AdamW
Learning rate	2×10^{-4}
LR schedule	Cosine decay
Warmup steps	10
Max sequence length	512 tokens
Precision	bfloat16
GPU	NVIDIA A100
Train / val / test	50 / 10 / 40

Table 6: Hyperparameters used for LoRA fine-tuning across all models.

Model	Provider	Access
<i>Closed-source</i>		
Gemini 2.5 Pro (Gemini Team, Google et al., 2025)	Google	API
Gemini 2.5 Flash (Gemini Team, Google et al., 2025)	Google	API
Claude Sonnet 4.6 (Anthropic, 2026)	Anthropic	API
Claude Haiku 4.5 (Anthropic, 2025)	Anthropic	API
GPT-5.4-mini (OpenAI, 2026)	OpenAI	API
GPT-5.4-nano (OpenAI, 2026)	OpenAI	API
<i>Open-source</i>		
Qwen3.5-9B (Qwen Team, 2026)	Alibaba	Open
Qwen3.5-27B (Qwen Team, 2026)	Alibaba	Open
Qwen3.5-35B-A3B (Qwen Team, 2026)	Alibaba	Open
Gemma-3-4B-it (Kamath et al., 2025)	Google	Open
Gemma-3-12B-it (Kamath et al., 2025)	Google	Open
InternVL3.5-2B (Wang et al., 2025)	OpenGVLab	Open
InternVL3.5-4B (Wang et al., 2025)	OpenGVLab	Open
InternVL3.5-8B (Wang et al., 2025)	OpenGVLab	Open
Llama 4 Maverick (Meta AI, 2025)	Meta	Open

Table 7: Vision-language models evaluated in our benchmark.

H Error-Classification Protocol

This section specifies how each (ground-truth, prediction) pair is assigned to exactly one of the fifteen categories in Tab. 8. No manual or LLM labeling enters at any stage; assignment is performed by a single deterministic, rule-based classifier, released with the code, and applied identically to all four VLM families and three conventional OCR baselines. Image-level events (script contamination, semantic hallucination, truncation, spurious extra content, numeric-only error) are counted once per image; word-level events come from images that produced predominantly correct Bengali but with localized errors, with one event per misaligned word.

Taxonomy construction. The taxonomy was developed inductively from a pilot error analysis of two representative systems (Gemini 2.5 Flash and Qwen3.5-9B) and cross-checked against prior Indic-OCR error typologies (Nayef et al., 2019; Gunna et al., 2022). We retained the conventional orthographic categories those typologies emphasize (matra substitution, consonant homophones, conjunct/hasanta errors, nasal diacritics, and compound orthographic edits) and added five categories specific to generative transcription that classical OCR typologies do not cover: script contamination, semantic hallucination, spurious extra content, truncation/no output, and numeric/temporal substitution. The category set was frozen before the full classification pass.

Determinism and cross-model consistency. The classifier receives only the pair (ground truth, prediction); it has no access to the source image, the prompt, or the identity of the model that produced the prediction. It therefore cannot assign the same (GT, prediction) input to different categories for different models, so cross-model consistency is guaranteed by construction rather than enforced by review. Given a fixed input the output is fully reproducible: there is no sampling, no data-fit threshold, and no human adjudication step.

Stage 1: image-level cascade. Each prediction is first tested for whole-output failure modes in a fixed priority order, before any word-level analysis. In order: (i) *truncation / no output* (empty string, refusal, or output length below 40% of the ground-truth length); (ii) *script contamination* (the

dominant Unicode block of the output is Devanagari, Arabic, or Latin despite a Bengali ground truth); (iii) *semantic hallucination* (the output is well-formed Bengali but shares under 25% of its characters with the ground truth); (iv) *spurious extra content* (the full ground-truth string is recovered but accompanied by unsolicited additional tokens, such as an appended translation or filler); and (v) *numeric / temporal error* (all non-numeric content matches after applying a numeric mask, and only Bengali digits differ). The first rule that fires assigns the category, and the event is counted once for the image. If none fires, the image is judged to contain predominantly correct Bengali with localized errors and is passed to Stage 2.

Stage 2: word-level operation counting. For images reaching Stage 2, the ground-truth and predicted strings are aligned word-by-word via a Levenshtein alignment. Each misaligned word is then aligned at the character level, and its edit operations (insertions, deletions, substitutions) are partitioned by the Unicode category of the characters involved: base consonants, dependent vowel signs (matra), hasanta/virama, nasal diacritics (chandra-bindu, anusvara, visarga), and digits. One event is emitted per misaligned word.

Hierarchical resolution to a single class. Each word-level event is resolved to exactly one category by a total, mutually exclusive decision hierarchy over the edit statistics, using the per-class thresholds listed in Tab. 8. Word-magnitude classes are tested first: a word is *lexical word substitution* when character overlap falls below 40% or the edit distance reaches 60% of the word length; *multi-character word corruption* when there are at least three grapheme edits with over 40% character overlap retained; and *visual grapheme misread* when a single visually-similar but linguistically-unrelated substitution occurs within an edit distance of two. If the edits are instead confined to a single orthographic Unicode category, the event resolves to the corresponding orthographic class: matra-only edits to *vowel-sign substitution*; hasanta or conjunct-ligature edits to *conjunct/hasanta error*; substitutions within a fixed homophone-class lookup (ন/ণ (*na/ṇa*), স/শ/ষ (*sa/śa/ṣa*), ব/ভ (*ba/bha*), ত/থ (*ta/tha*), and similar) to *consonant homophone*; nasal-diacritic-only edits to *nasal diacritic error*; and edits spanning two or more orthographic categories (with no unrelated consonant substitution) to *compound orthographic*. Pairs that differ only

after NFC normalization or in punctuation/quote glyphs resolve to *unicode normalization*. Because the hierarchy is exhaustive and its branches are disjoint, every event maps to exactly one of the fifteen categories.

Counting convention. Image-level events (script contamination, semantic hallucination, truncation/no output, spurious extra content, and numeric/temporal error) are counted once per image; word-level events are counted once per misaligned word. The per-model percentages in Tab. 2 are the share of each category among all events attributed to a given system, and the visual-to-orthographic ratios reported in § 6 are computed from these shares.

I Error Analysis

Three examples of each of the error categories specified in Tab. 2 can be found in Tab. 9 and Tab. 10.

J Error Taxonomy

The definitions of each of the error categories specified in Tab. 2 can be found in Tab. 8.

Error type	Description
Lexical word substitution	Whole word replaced by an unrelated lexical item (char overlap < 40% or edit distance $\geq$ 60% of length). Visually-driven recognition failure at the word level, e.g. ফ্লাটটি(<i>phlyāṭṭi</i>)→ফ্যাক্টরি(<i>phyāktari</i>), কপাট(<i>kapāṭa</i>)→লোহ(<i>loha</i>), বৈষম্য(<i>baiṣamya</i>)→লোন(<i>lona</i>).
Visual grapheme misread	Single-character substitution between visually-similar but linguistically-unrelated graphemes (edit distance $\leq$ 2 over a word). Reflects feature-level vision-encoder error, e.g. মেধা(<i>medhā</i>)→মেবা(<i>mebā</i>), চুইঝাল(<i>cuijhāla</i>)→চুইমাল(<i>cuimāla</i>).
Word-boundary segmentation	Adjacent words concatenated (এনা(<i>enā</i>)→এন(<i>ena</i>) এ(<i>e</i>)) or a single token spuriously split (পাটকল(<i>pāṭakala</i>)→পটল কল(<i>paṭala kala</i>)); also covers whitespace-only normalization. Inflates WER but leaves the character sequence largely intact.
Multi-character word corruption	Word recognizable in shape but with $\geq$ 3 grapheme edits and > 40% character overlap (e.g. পাঞ্জেরী(<i>pāñjerī</i>)→পান্তরী(<i>pāntarī</i>), স্বাধীনতা(<i>svādhīnatā</i>)→সাবধানতা(<i>sābadhanatā</i>). Partial visual recovery without lexical replacement.
Vowel-sign (matra) substitution	Only dependent vowel signs differ between GT and prediction (ি/ী (<i>i/ī</i>), ু/ূ (<i>u/ū</i>), ে/ৈ (<i>e/ai</i>), ো/ৌ (<i>o/au</i>)). Base consonants and conjuncts are preserved.
Script contamination	Output drifts into Devanagari, Arabic, or dominant Latin transliteration despite Bengali input. Distinct cross-lingual failure mode.
Unicode normalization	Strings differ only after NFC normalization, or differ in punctuation/quote glyphs (e.g. " vs “, - vs –). Penalized by CER/WER but visually identical.
Consonant homophone	Substitution within a known orthographic-homophone class (ন/ণ (<i>na/ṇa</i>), স/শ/ষ (<i>sa/śa/ṣa</i>), ব/ভ(<i>ba/bha</i>), ত/থ (<i>ta/tha</i>), etc.). Pure spelling-knowledge failure with the right phonetic target.
Conjunct / hasanta error	Hasanta (়) dropped or spuriously inserted, conjunct ligature mis-segmented (স্ব(<i>sva</i>)→সব(<i>saba</i>), ক্ষ(<i>kṣa</i>)→খ(<i>kha</i>), ব্র(<i>bra</i>)→বর(<i>bara</i>)). Leaves the base consonant identities ambiguous.
Compound orthographic	Multiple orthographic-type errors (matra + nasal + homophone) co-occur in a single word, with no unrelated consonant substitution.
Semantic hallucination	Whole-image output is fluent Bengali but shares < 25% characters with GT (language-prior takeover, e.g. ছাত্র ইউনিয়ন(<i>chātra iuniyana</i>)→শুভ জন্মদিন(<i>śubha janmadina</i>)).
Numeric / temporal error	All non-numeric text correct; only Bengali numerals are wrong (১৪৩৩(<i>1433</i>)→১৪৩০(<i>1430</i>), ২০২৫(<i>2025</i>)→২০২৩(<i>2023</i>)).
Nasal diacritic error	Chandrabindu (ঁ), anusvara (ং), or visarga (ঃ) confused or dropped, no other change.
Truncation / no output	Empty output, refusal, or length below 40% of ground truth.
Spurious extra content	GT text correctly present plus unsolicited translation or filler.

Table 8: Definitions of the fifteen error categories in the BANGLAWILD taxonomy (see Table 2 for per-model rates). Bengali examples are shown with ALA-LC romanization.

Error type	M	Ground truth	Model prediction
Lexical word substitution <i>Whole word replaced by an unrelated lexical item.</i>	G	কল্যাণ রাষ্ট্র চাই <i>kalyāṇa rāṣṭra cāi</i>	কল্যাণপুর <i>kalyāṇapura</i>
	G	সব শিশুরা হাদি হবে! <i>saba śiśurā hādi habe!</i>	সব শ্রমিক এক হবে! <i>saba śramika eka habe!</i>
	Q	অসাম্প্রদায়িক বাংলাদেশ <i>asāmpradāyika bāṁlādeśa</i>	আমাদের বাংলাদেশ <i>āmādera bāṁlādeśa</i>
Visual grapheme misread <i>Single-character substitution between similar graphemes.</i>	G	গদি ছাড় <i>gadi chāra</i>	সদি ছাড় <i>sadi chāra</i>
	G	জান দেবো জুলাই দেবোনা <i>jāna debō jūlāi debonā</i>	জান দেবো জুলুম দেবো না <i>jāna debō jūluma debō nā</i>
	Q	ধর্ম যার যার দেশটা সবার <i>dharma yāra yāra deśaṭā sabāra</i>	ধর্ম যার যার দেশটা আমার <i>dharma yāra yāra deśaṭā āmāra</i>
Word-boundary segmentation <i>Adjacent words concatenated or a single token split.</i>	G	সমঅধিকার-সমমর্যাদা নিশ্চিত কর <i>samaadhikāra-samamaryādā niścita kara</i>	সম অধিকার- সম মর্যাদা নিশ্চিত কর <i>sama adhikāra- sama maryādā niścita kara</i>
	G	অটোপরিবহন <i>aṭōparibahana</i>	অটো পরিবহন <i>aṭō paribahana</i>
	G	রাষ্ট্রীয় ও দলীয় সম্মান মুক্ত... <i>rāṣṭrīyāo dalīya santrāsa mukta...</i>	রাষ্ট্রীয় ও দলীয় সম্মান মুক্ত... <i>rāṣṭrīyā o dalīya santrāsa mukta...</i>
Multi-character word corruption <i>Word recognizable but with ≥ 3 grapheme edits.</i>	G	কালচারাল ফ্যাসিজম নিপাত যাক <i>kālacārāla phyāsijama nipāta yāka</i>	কালচারাল ফ্যাসিবাদ নিপাত যাক <i>kālacārāla phyāsibāda nipāta yāka</i>
	G	দাবি...চাই নিরদলীয় সরকার <i>dābi...cāi nirdalaīya sarakāra</i>	দাবি...চাই নিরালা সরকার <i>dābi...cāi nirālā sarakāra</i>
	Q	স্বৈরাচার মুক্ত বাংলাদেশ <i>sbairācāra mukta bāṁlādeśa</i>	মেরচার মুক্ত বাংলাদেশ <i>meracāra mukta bāṁlādeśa</i>
Vowel-sign substitution <i>Only dependent vowel signs differ.</i>	G	শহিদ মিনার আবাসিক এলাকা <i>śahida mināra ābāsika elākā</i>	শহীদ মিনার আবাসিক এলাকা <i>śahida mināra ābāsika elākā</i>
	G	লাখো...কেনা দেশটা কারো <i>lākho...kenā deśaṭā karo</i>	লাখো...কেনা দেশটি কারো <i>lākho...kenā deśaṭi karo</i>
	Q	পাল্টা আঘাত <i>pālṭā āghāta</i>	পল্টা আঘাত <i>palṭā āghāta</i>
Script contamination <i>Output drifts into Devanagari, Arabic, or Latin.</i>	G	যুদ্ধ ৯ শান্তি <i>yuddha o śānti</i>	WAR VS PEACE
	G	রাখতে <i>rākhate</i>	SEMPRE
	Q	হাল ছেড়োনা...কণ্ঠ ছাড়ো জোরে <i>hāla cheronā...kanṭha chāro jore</i>	
Unicode normalization <i>Strings differ only after NFC or in punctuation glyphs.</i>	G	ধর্মীয় হিংসা নয়...সাম্প্রদায়িক... <i>dharmīya naya...sāmpradāyika...</i>	ধর্মীয় হিংসা নয়...সাম্প্রদায়িক... <i>dharmīya naya...sāmpradāyika...</i>
	G	নাটক কম করো পিও! <i>nātaka kama karo pio!</i>	নাটক কম করো পিও! <i>nātaka kama karo pio! (NFC-only)</i>
	Q	ধর্মীয় হিংসা নয়... সাম্প্রদায়িক... <i>dharmīya himsā naya... sām-pradāyika...</i>	ধর্মীয় হিংসা নয়... সাম্প্রদায়িক... <i>dharmīya himsā naya... sām-pradāyika...</i>

Table 9: Three real examples per error category (part 1 of 2), drawn from the model outputs of **G**=Gemini-2.5-Flash and **Q**=Qwen3.5-9B. For word-level categories the affected word is underlined; for image-level categories the full transcription is shown. Each cell shows Bengali script (top) and ALA-LC romanization (bottom, *italic*); differing characters are underlined. Romanization follows ALA-LC 2017 ([American Library Association and Library of Congress, 2017](#)). *Continued in Table 10.*

Error type	M	Ground truth	Model prediction
Consonant homophone <i>Within a known homophone class (ন/ণ, স/শ/ষ...).</i>	G	সাধারণ সম্পাদক <i>sādhārana sampādaka</i>	সাধারণ সম্পাদক <i>sādhārana sampādaka</i>
	G	রক্ত গরম মাথা ঠাণ্ডা! <i>rakta garama mātā thāṇḍā!</i>	রক্ত গরম মাথা ঠাণ্ডা! <i>rakta garama mātā thāṇḍā!</i>
	Q	স্লোগান তোমার বাংলাদেশ সবার <i>slōgāna tomāra bāṁlādeśa sabāra</i>	স্লোগান তোমার বাংলাদেশ আমার <i>slōgāna tomāra bāṁlādeśa āmāra</i>
Conjunct / hasanta error <i>Hasanta dropped or conjunct mis-segmented.</i>	G	আল্লাহ্ ভরসা <i>āllāh bharasā</i>	আল্লাহ্ ভরসা <i>āllāha bharasā</i>
	G	স্বাধীনতা <i>sbādhīnatā</i>	সাবধানতা <i>sābadhānatā</i>
	Q	আমাদের স্বাধীন দেশে... <i>āmādera sbādhīna deśe...</i>	আমাদের সুবিধীন দেশে... <i>āmādera subidhīna deśe...</i>
Compound orthographic <i>Multiple orthographic-type errors in one word.</i>	G	তারুণ্যের প্রথম ভোট <i>tārūnyera prathama bhoṭa</i>	তারুণ্যের প্রথম ভোট <i>tārūnyera prathama bhoṭa</i>
	G	মডার্ন স্টেশনারী <i>maḍārṇa ṣṭeśanārī</i>	মডার্ন স্টেশনারী <i>maḍārṇa ṣṭeśanārī</i>
	Q	২৪ এর... তরুণ যেন... <i>24 era...tarūṇa yena...</i>	২৪ এর... তরুন যেন... <i>24 era...taruna yena...</i>
Semantic hallucination <i>Fluent Bengali, < 25% char overlap with GT.</i>	G	৩৬শে জুলাই <i>36śe julāi</i>	ভোট দিন <i>bhoṭa dina</i>
	G	ওয়েবসাইট লাগবে? <i>oyebosāiṭa lāgabe?</i>	ধূমপান? <i>dhūmapāna?</i>
	Q	স্বাদ আর বিশুদ্ধতায় সেরা <i>sbāda āra biśuddhatāya serā</i>	মাদার <i>mādāra</i>
Numeric / temporal error <i>Surrounding text correct; numerals wrong.</i>	G	শুভ নববর্ষ ১৪৩৩ <i>śubha nababarṣa 1433</i>	শুভ নববর্ষ ১৪৩০ <i>śubha nababarṣa 1430</i>
	G	মাসব্যাপী...উৎসব ২০২৫ <i>māsabyāpī...utsaba 2025</i>	মাসব্যাপী...উৎসব ২০২৩ <i>māsabyāpī...utsaba 2023</i>
	Q	মেরামতের ৩১ দফা <i>merāmatēra 31 daphā</i>	মেরামতের ৬১ দফা <i>merāmatēra 61 daphā</i>
Nasal diacritic error <i>Chandrabindu, anusvara, or visarga differs.</i>	G	ফুল ছেঁড়া নিষেধ সুজন <i>phula cherā niṣedha sujana</i>	ফুল ছেঁড়া নিষেধ সুজন <i>phula chem ṛā niṣedha sujana</i>
	G	বৈষম্যের ঠাই নাই <i>baiṣamyera thām i nāi</i>	বেগমোর ঠাই নাই <i>begamora thāi nāi</i>
	G	আড়ং ন্যাচারাল চিনিগুড়া চাল <i>āraṁ nyācārāla ciniguṛā cāla</i>	আড়ং ন্যাচারাল চিনিগুড়া চাল <i>āraṁ nyācārāla cinigum ṛā cāla</i>
Truncation / no output <i>Empty refusal or length < 40% of GT.</i>	G	বল বীর চির উন্নত মম শির <i>bala bīra cira unnata mama śira</i>	জন্মভূমির <i>janmabhūmira</i>
	Q	শহীদের রক্ত হবে নাকো ব্যর্থ। <i>śahīdera rakta habe nāko byartha.</i>	বঙ্গবন্ধু <i>baṅgabandhu</i>
	Q	চির উন্নত মম শির <i>cira unnata mama śira</i>	চিরতরে <i>ciratare</i>
Spurious extra content <i>GT text correct plus unsolicited additions.</i>	G	কবির পরিবহন <i>kabira paribahana</i>	কবির পরিবহন আশরাফুল ও আলামিন <i>kabira āsarāphula o ālāmīna paribahana</i>
	Q	কাজী অফিস <i>kājī aphisa</i>	৫.১১ কাজী অফিস <i>5.11 kājī aphisa</i>
	Q	আশরাফ আলী <i>āsarāpha ālī</i>	আশরাফ আলী দা কুশা <i>āsarāpha ālī dā kuścā</i>

Table 10: Three real examples per error category (part 2 of 2; see Table 9 for part 1), drawn from the model outputs of G=Gemini-2.5-Flash and Q=Qwen3.5-9B. For word-level categories the affected word is underlined; for image-level categories the full transcription is shown. Each cell shows Bengali script (top) and ALA-LC romanization (bottom, *italic*); differing characters are underlined. Romanization follows ALA-LC 2017 (American Library Association and Library of Congress, 2017).

Table 11: Model benchmarking results (CER %) on the BANGLAWILD test set across **Content** and **Context** attribute categories under three prompting strategies: **P1** (Baseline English), **P2** (Baseline Bengali), and **P3** (Structured English). **Cyan** highlights the lowest CER per column; **bold** marks the best-performing model per group.

Model	Content																	
	Advertisement			Colloquial			Creative			Instructional			Political			Religious		
	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
Claude Haiku 4.5	68.80	68.12	62.37	183.96	146.16	140.16	100.68	93.62	102.46	63.35	62.91	65.55	120.96	102.08	136.42	46.15	45.91	45.94
Claude Sonnet 4.6	14.88	12.33	12.44	30.26	34.13	27.40	24.14	21.77	19.58	17.52	11.85	10.81	41.07	25.28	27.05	13.81	12.79	13.17
GPT 5.4-mini	42.31	43.79	43.55	66.71	81.33	65.55	63.65	67.69	62.30	39.87	48.19	37.85	76.29	78.55	71.49	31.14	32.68	31.70
GPT 5.4-nano	208.31	168.39	110.27	381.47	258.86	128.38	273.31	207.62	121.74	213.33	187.52	109.71	338.54	246.69	127.00	117.75	123.29	94.27
Gemini 2.5 Flash	12.42	12.32	12.32	21.69	27.91	20.82	13.39	13.43	13.58	8.76	8.89	8.76	17.42	21.01	17.70	11.94	14.26	11.02
Gemini 2.5 Pro	21.05	18.40	23.54	27.11	23.59	23.98	28.79	25.05	22.72	27.83	17.71	21.84	26.43	24.29	27.42	24.04	21.09	22.28
Gemma-3-12B-it	70.07	97.92	47.92	60.06	62.65	61.30	63.90	51.76	46.97	25.65	28.58	62.15	57.35	67.14	63.88	72.68	56.81	45.88
Gemma-3-4B-it	122.19	89.70	68.22	83.34	88.13	77.88	165.02	116.95	70.39	43.33	44.85	43.49	160.29	178.83	80.65	142.01	146.18	78.88
InternVL3.5-2B	114.03	168.66	121.81	158.72	227.48	132.95	147.21	177.66	142.14	136.02	178.41	144.69	153.90	194.03	155.85	112.81	144.18	131.49
InternVL3.5-4B	114.15	156.77	112.18	171.74	233.29	110.87	112.74	148.57	116.33	117.14	148.06	120.74	117.03	159.41	116.29	111.53	130.02	120.70
InternVL3.5-8B	108.46	116.59	103.10	113.78	141.16	106.58	123.25	116.43	101.60	100.69	101.04	112.01	135.60	122.04	105.51	104.73	103.91	103.35
Llama 4 Maverick	32.89	36.31	36.08	64.97	56.43	62.02	38.14	39.48	38.97	23.23	30.32	22.94	45.21	44.76	46.36	23.65	23.79	20.70
Qwen3.5-9B	16.13	15.80	15.69	31.27	30.01	29.05	22.21	22.34	23.02	13.90	13.80	13.46	27.71	28.90	26.95	14.03	14.50	13.99
Qwen3.5-27B	18.19	18.02	18.54	39.10	34.60	37.43	26.09	27.10	25.62	15.85	15.00	14.31	33.15	32.32	33.00	15.87	15.77	16.13
Qwen3.5-35B-A3B	17.27	15.92	16.68	34.72	34.37	31.46	21.79	21.54	21.57	13.90	13.74	12.83	28.33	27.21	27.81	14.97	13.61	14.20

Model	Context											
	Packaging			Printed Material			Signboard			Vehicle		
	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
Claude Haiku 4.5	72.92	67.97	67.15	51.95	47.45	52.55	79.08	79.41	63.62	126.45	120.66	110.74
Claude Sonnet 4.6	16.35	12.73	14.44	11.01	9.79	9.21	18.80	12.47	12.77	27.93	25.97	24.31
GPT 5.4-mini	50.51	50.49	52.00	33.30	34.58	31.25	47.70	48.02	45.15	56.95	63.44	58.60
GPT 5.4-nano	198.38	128.76	109.40	190.13	159.39	107.46	236.35	178.03	108.37	202.15	184.12	121.14
Gemini 2.5 Flash	10.26	9.88	10.57	7.70	7.31	7.94	13.69	13.62	12.92	30.35	40.11	26.91
Gemini 2.5 Pro	16.53	10.66	13.93	20.97	14.73	24.23	25.63	19.58	24.36	30.44	27.85	27.33
Gemma-3-12B-it	47.18	95.76	47.07	31.50	31.81	31.67	113.64	43.95	42.86	111.09	65.62	60.68
Gemma-3-4B-it	55.33	66.02	58.82	52.45	46.87	44.94	57.48	197.39	75.11	111.57	83.24	84.07
InternVL3.5-2B	111.62	172.86	112.95	114.79	166.29	126.73	124.04	154.94	118.98	137.75	238.98	134.85
InternVL3.5-4B	109.66	153.91	101.33	105.30	138.73	108.20	118.65	169.91	112.56	172.73	217.80	120.47
InternVL3.5-8B	103.35	113.62	99.44	107.37	108.85	98.81	108.32	117.44	103.83	134.94	141.34	109.98
Llama 4 Maverick	26.83	24.99	27.83	22.85	21.71	21.79	34.40	44.02	32.96	56.62	51.14	60.85
Qwen3.5-9B	19.59	18.35	18.21	13.58	13.98	12.48	17.05	16.05	16.41	29.29	28.50	29.73
Qwen3.5-27B	23.70	22.99	26.65	15.25	14.86	15.08	19.14	19.31	18.85	36.61	31.21	34.46
Qwen3.5-35B-A3B	22.61	21.08	21.46	14.20	12.19	13.49	17.55	15.70	16.69	34.19	32.83	32.44

P1: Baseline English **P2:** Baseline Bengali **P3:** Structured English Metric: CER (%), lower is better).

Table 12: Model benchmarking results (CER %) on the BANGLAWILD test set across image attribute categories under three prompting strategies: **P1** (Baseline English), **P2** (Baseline Bengali), and **P3** (Structured English). **Cyan** highlights the lowest CER per column; **bold** marks the best-performing model per group.

Model	Background Complexity									Curved Text								
	Clean			Cluttered			Moderate			No			Yes					
	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3			
Claude Haiku 4.5	88.17	78.48	88.02	123.28	138.19	79.69	93.46	85.77	96.79	84.79	79.11	83.34	172.35	126.27	166.30			
Claude Sonnet 4.6	22.11	17.10	17.64	27.79	23.03	23.59	28.32	25.82	17.96	19.51	15.98	16.73	75.55	54.20	35.66			
GPT 5.4-mini	51.82	54.98	50.81	64.65	67.58	66.19	59.83	65.17	58.52	51.46	53.99	50.59	81.52	97.87	79.81			
GPT 5.4-nano	237.46	182.33	111.78	304.41	287.59	130.53	288.69	227.07	129.17	242.06	190.91	113.94	317.48	219.97	127.49			
Gemini 2.5 Flash	13.46	15.21	13.30	24.73	23.80	24.98	14.22	15.13	14.04	13.07	14.66	12.90	28.69	29.22	28.87			
Gemini 2.5 Pro	24.03	20.92	23.15	37.05	34.76	32.97	25.76	21.58	25.87	24.19	20.67	23.41	34.46	35.77	31.66			
Gemma-3-12B-it	61.18	59.35	48.11	64.50	75.73	86.85	81.78	144.94	67.67	56.47	70.13	51.34	170.96	79.85	66.94			
Gemma-3-4B-it	120.13	106.48	71.11	192.50	134.49	92.62	211.08	201.81	73.24	136.25	120.63	70.88	114.16	105.35	94.05			
InternVL3.5-2B	130.91	177.94	132.36	138.47	162.67	171.23	142.69	173.64	147.52	131.56	174.15	136.75	149.52	213.42	126.03			
InternVL3.5-4B	117.35	155.12	113.05	114.57	167.61	116.15	127.14	164.96	130.81	117.57	154.92	115.35	130.93	185.80	116.15			
InternVL3.5-8B	117.25	117.33	104.91	138.78	113.43	100.65	102.07	115.06	99.22	114.88	115.64	103.04	137.83	134.61	118.02			
Llama 4 Maverick	35.02	36.07	34.81	58.38	65.36	57.64	41.85	40.44	49.82	35.30	36.03	35.89	60.82	66.01	64.48			
Qwen3.5-9B	19.69	19.78	19.49	27.94	27.94	26.83	22.16	22.62	21.14	18.88	19.07	18.57	41.98	41.29	41.27			
Qwen3.5-27B	23.35	22.44	23.15	31.99	36.92	29.55	24.29	25.15	25.27	22.23	21.77	21.99	47.46	47.66	48.44			
Qwen3.5-35B-A3B	20.83	19.79	20.12	26.52	26.12	26.12	21.65	20.99	20.50	19.54	18.73	18.89	45.06	41.87	42.89			

Model	Font Style									Occlusion								
	Artistic			Handwritten			Printed			Heavy			None			Partial		
	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
Claude Haiku 4.5	82.51	66.33	86.93	104.05	96.98	102.58	66.08	62.65	57.59	258.78	211.35	78.38	85.97	76.21	85.88	99.39	99.03	100.64
Claude Sonnet 4.6	25.57	20.93	18.14	25.82	20.53	20.46	14.40	11.10	11.83	23.65	24.46	22.57	21.46	17.36	17.24	29.88	22.56	20.58
GPT 5.4-mini	44.05	48.16	41.89	62.83	67.15	62.22	40.30	40.91	39.84	60.14	60.54	64.19	51.66	54.56	50.65	60.06	65.75	59.21
GPT 5.4-nano	190.65	161.74	102.22	289.28	224.78	123.58	202.08	147.55	106.46	293.11	211.89	120.27	237.21	185.92	111.78	283.44	219.27	126.68
Gemini 2.5 Flash	14.85	18.32	15.03	15.17	16.70	14.95	10.73	10.31	10.43	21.08	21.08	22.84	12.87	14.40	12.95	18.56	20.11	17.40
Gemini 2.5 Pro	28.45	27.24	26.41	25.39	21.96	23.92	20.02	15.37	21.58	47.03	42.70	28.24	23.56	20.99	23.06	28.81	23.14	27.25
Gemma-3-12B-it	64.60	73.82	64.82	74.20	77.01	53.18	38.73	52.89	38.04	63.51	63.51	61.08	46.81	61.11	49.95	132.27	109.77	61.46
Gemma-3-4B-it	179.72	114.05	84.55	150.59	149.67	75.27	53.03	53.87	53.50	74.05	68.92	75.54	129.52	122.34	72.23	159.26	111.54	72.82
InternVL3.5-2B	127.66	156.82	133.85	143.42	189.20	143.73	112.35	166.65	120.00	131.22	143.11	168.92	131.06	176.25	134.08	139.50	180.29	142.17
InternVL3.5-4B	112.69	144.17	118.51	123.93	164.45	118.65	112.26	151.66	104.61	119.32	182.57	122.03	117.50	154.27	113.07	122.13	166.16	124.38
InternVL3.5-8B	111.68	109.26	105.74	123.93	121.03	105.06	102.90	114.49	98.81	94.59	107.84	102.43	112.57	117.41	104.40	132.74	115.17	102.53
Llama 4 Maverick	33.66	33.52	36.11	42.20	42.49	42.99	27.77	31.66	26.89	65.00	96.49	67.03	33.57	35.40	34.66	49.00	45.12	48.51
Qwen3.5-9B	19.12	18.43	18.25	22.74	23.29	22.90	16.02	15.97	15.03	25.68	27.92	22.16	19.56	19.58	19.27	23.39	23.86	23.03
Qwen3.5-27B	22.45	20.93	22.29	26.69	26.69	26.28	18.57	18.25	19.01	28.11	27.97	26.49	23.04	22.23	22.86	26.94	28.11	26.94
Qwen3.5-35B-A3B	19.69	18.78	18.75	23.33	22.70	22.66	17.61	15.81	16.86	24.73	24.32	23.65	20.30	19.05	19.49	24.59	24.76	24.13

Ground Truth	Corrected Version	Qwen3.5-9B	Qwen3.5-35B-A3B	Claude Sonnet 4.6	Gemini 2.5 Flash
অনুমোতিতে <i>anumōtite</i>	অনুমতিতে <i>anumatite</i>	অনুমোদিতে <i>anumodite</i>	অনুমোতিতে <i>anumōtite</i>	অনুমতিতে <i>anumatite</i>	অনুমতিতে <i>anumatite</i>
ধানমণ্ডি <i>dhānamaṇḍi</i>	ধানমন্ডি <i>dhānamaṇḍi</i>	ধানমন্ডি <i>dhānamaṇḍi</i>	ধানমন্ডি <i>dhānamaṇḍi</i>	ধানমন্ডি <i>dhānamaṇḍi</i>	ধলমাটি <i>dhalamāṭi</i> (halluc.)
বলবীর <i>balabīr</i>	বল বীর <i>bala_bīr</i>	বলশীর্ <i>balaśīr</i>	বন বিক্র <i>banabikra</i>	জালজীবা (Hindi) <i>jalajīva</i> (Hindi)	বলবীর <i>balabīr</i> ✓
সুবিদা <i>subidā</i>	সুবিভা <i>subibhā</i>	সুবিভা <i>subibhā</i>	সুবিদা <i>subidā</i>	সুবিভা <i>subibhā</i>	সুবিধা <i>subidhā</i>
পোস্টার <i>pōstār</i>	পোস্টার <i>pōstār</i>	পোস্টার <i>pōstār</i>	পোস্টার <i>pōstār</i>	পোস্টার <i>pōstār</i>	পোস্টার <i>pōstār</i>
ম্যাট্রামনি <i>myātramni</i>	ম্যাট্রামনি <i>myātraimni</i>	ম্যাট্রামনি <i>myātramni</i>	ম্যাট্রামনি <i>myātramni</i>	ম্যাট্রামনি <i>myātraimni</i>	ম্যাজিকমণি <i>myājikamoṇi</i> (halluc.)
হর্ন <i>harṇ</i>	হর্ন <i>harṇ</i>	জন <i>jana</i> (halluc.)	তর্ক <i>tarka</i> (halluc.)	হর্ন <i>harṇ</i>	শুভ নববর্ষ <i>śubha nababarṣa</i> (halluc.)
স্বাধীন <i>sbādhīna</i>	স্বাধীন <i>sbādhīna</i>	স্বাধীন <i>sbādhīna</i>	অসাধারণ <i>asādhāraṇa</i> (halluc.)	স্বাধীন <i>sbādhīna</i>	স্বাধীন <i>sbādhīna</i>
জীত <i>jīta</i>	জিত <i>jīta</i>	জীবিত <i>jībita</i>	জীব <i>jība</i>	উত্বিত <i>utita</i> (halluc.)	জীত <i>jīta</i> ✓
আষাড়েও <i>āṣāṛeo</i>	আষাড়েও <i>āṣāḍheo</i>	ফারগুনেই <i>phāraguneī</i> (halluc.)	আসাদেও <i>āsādeo</i>	ফাগুনেই <i>phāguneī</i> (halluc.)	ফাগুনেই <i>phāguni</i> (halluc.)

Table 13: Word-level mismatch analysis between ground truth and corrected version, with corresponding model outputs for the same word. Each row shows the Bengali script (top) and ALA-LC romanization (bottom, *italic*); the differing character(s) are underlined. Key orthographic distinctions invisible to non-Bengali readers: *ṇ* (ঞ, retroflex nasal) vs. *n* (ন, dental nasal); long vowel *ī* (ী) vs. short *i* (ি); long vowel *ā* (া) vs. implicit short *a*; retroflex sibilant *ṣ* (ষ) vs. dental *s* (স); and retroflex *r* (ড়) vs. aspirated *ḍh* (ঢ়). These substitutions are counted as character errors in CER. A ✓ denotes output matching the ground truth. (halluc.) marks outputs with no lexical overlap with the source word. Romanization follows ALA-LC 2017 ([American Library Association and Library of Congress, 2017](#)).

Table 14: Per-category OCR performance for **Gemini-2.5-Flash** and **Qwen3-9B** across all 24 (**Context** $\times$ **Content**) groups, under the three prompting regimes: **P1** (English baseline), **P2** (Bengali baseline), and **P3** (English structured). All metrics are reported as percentages; lower is better ($\downarrow$). For each (model, metric) triple, the best of {P1, P2, P3} per row is **bold**. Rows are sorted by Gemini’s P2 CER, easiest at top. The bottom row reports the macro-average across all 24 groups. Macro-average is computed over the 24 observed (Context, Content) pairs shown; 6 of the 30 possible combinations contain no images and are excluded.

Context	Content	Gemini-2.5-Flash									Qwen3-9B								
		CER $\downarrow$			WER $\downarrow$			NED $\downarrow$			CER $\downarrow$			WER $\downarrow$			NED $\downarrow$		
		P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3
Printed Material	Instructional	2.33	2.33	2.52	6.91	6.91	7.61	2.30	2.30	2.47	3.78	5.16	3.78	11.80	11.80	11.80	3.61	4.53	3.61
Signboard	Instructional	5.81	6.10	5.94	15.67	17.83	18.17	5.01	5.29	5.15	9.57	9.74	9.20	27.05	27.39	27.05	8.59	8.78	8.25
Printed Material	Advertisement	10.63	9.93	10.73	23.62	24.89	24.79	9.92	9.00	9.77	15.73	23.40	15.99	36.65	44.05	35.52	14.99	14.22	14.99
Wall	Religious	10.28	10.23	10.94	31.33	31.76	32.12	9.78	9.76	10.35	14.39	16.53	14.73	41.55	43.49	41.97	13.92	14.99	14.28
Printed Material	Creative	16.95	11.91	16.95	62.66	35.00	62.66	13.35	8.96	13.35	7.43	8.35	7.43	39.58	47.91	39.58	7.20	8.16	7.20
Printed Material	Religious	14.56	12.29	15.26	37.01	36.26	40.08	12.79	10.49	13.36	22.44	21.91	21.36	49.43	48.06	47.00	21.10	20.87	19.98
Wall	Advertisement	13.48	13.66	14.24	33.13	34.34	34.75	13.06	13.17	13.77	13.22	12.67	12.76	39.51	38.08	38.91	12.41	12.07	12.42
Packaging	Colloquial	14.55	14.55	16.36	77.78	77.78	88.89	14.55	14.55	16.07	30.91	36.36	32.73	77.78	88.89	88.89	30.91	36.36	32.73
Packaging	Advertisement	16.20	14.70	16.06	33.73	33.18	33.62	14.29	12.63	14.34	25.16	22.76	22.75	52.31	51.26	50.02	22.99	20.41	21.77
Wall	Creative	15.11	15.41	15.43	28.13	28.49	28.55	14.52	14.37	14.57	25.41	25.49	26.45	48.94	49.29	49.24	24.04	24.30	23.70
Printed Material	Colloquial	19.62	16.27	19.68	37.62	36.15	39.09	17.08	14.88	17.70	38.35	34.73	35.58	61.38	60.28	62.85	33.36	32.66	34.52
Signboard	Political	15.29	16.35	12.18	43.39	42.32	36.07	14.84	15.78	11.71	18.26	19.33	21.61	52.54	44.50	58.21	17.78	18.45	21.09
Printed Material	Political	16.76	17.59	16.76	20.83	20.83	20.83	14.12	15.42	14.12	24.02	24.02	21.61	37.10	37.10	41.27	23.99	21.49	21.34
Wall	Instructional	18.79	18.01	18.26	32.37	32.39	33.01	13.48	12.98	13.37	27.41	26.26	28.39	47.35	44.84	44.77	25.56	23.80	25.97
Signboard	Religious	16.13	18.32	16.22	37.76	37.25	38.12	16.10	17.63	16.09	20.88	20.05	22.41	46.45	46.55	44.01	18.88	19.08	19.44
Vehicle	Instructional	19.53	21.51	18.89	41.67	41.67	38.89	19.42	20.56	18.78	18.76	17.93	16.53	50.83	43.19	48.06	18.61	17.70	16.33
Signboard	Advertisement	22.53	21.81	20.32	40.26	40.61	39.90	17.57	16.88	16.73	25.86	25.20	24.81	50.31	50.54	52.00	22.42	21.45	21.94
Wall	Political	22.61	24.03	22.49	39.18	41.93	39.07	20.09	19.90	19.97	34.30	39.40	33.74	61.05	65.41	60.67	31.02	30.40	30.94
Signboard	Colloquial	27.45	26.72	26.58	51.30	50.56	50.56	26.36	25.81	26.10	33.01	28.14	32.84	58.46	53.52	59.38	31.00	26.67	32.19
Vehicle	Advertisement	31.08	32.11	29.26	60.25	59.34	57.44	26.46	25.81	25.54	31.76	30.17	33.41	65.69	62.35	65.23	28.11	27.41	28.97
Wall	Colloquial	33.08	34.91	30.05	52.65	54.16	46.42	27.09	25.91	26.03	40.36	43.68	37.93	65.48	69.00	65.58	37.68	35.83	37.03
Vehicle	Religious	31.32	44.65	23.11	47.56	48.46	48.68	21.74	22.93	21.48	26.96	24.88	26.90	60.82	62.86	61.73	25.87	23.50	25.81
Vehicle	Colloquial	33.03	47.48	33.64	65.14	77.64	62.82	29.55	30.76	30.74	46.11	61.64	45.00	76.16	91.26	73.24	39.90	36.08	39.69
Vehicle	Creative	60.38	48.96	48.22	59.58	48.33	50.83	29.35	27.78	27.54	36.64	32.85	36.64	64.17	61.67	64.17	35.17	29.88	35.17

Model	Prompt	Exact match (%)		
		Verbatim	Standard	Neither
Gemini 2.5 Flash	P1	5.03	24.85	70.12
	P2	3.02	26.03	70.95
	P3	3.38	23.40	73.22
Gemini 2.5 Pro	P1	5.47	24.37	70.16
	P2	9.11	19.48	71.41
	P3	8.28	16.15	75.57
Claude Sonnet 4.6	P1	8.00	8.04	83.96
	P2	4.64	11.38	83.98
	P3	7.37	12.02	80.61
Qwen3.5-35B-A3B	P1	4.15	6.79	89.06
	P2	3.11	7.92	88.97
	P3	4.15	8.15	87.70
Llama 4 Maverick	P1	4.53	6.62	90.79
	P2	4.68	5.52	89.65
	P3	2.22	4.98	91.00
Qwen3.5-9B	P1	3.69	4.41	92.07
	P2	3.63	4.44	91.83
	P3	4.11	5.43	90.39
Qwen3.5-27B	P1	0.95	4.18	94.87
	P2	0.95	3.63	95.42
	P3	0.95	3.63	95.42
GPT-5.4-mini	P1	0.95	0.00	99.05
	P2	0.95	1.90	97.15
	P3	0.95	3.80	95.25
Gemma-3-12B-it	P1-P3	0.00	0.00	100.00
Claude Haiku 4.5	P1-P3	0.00	0.00	100.00
GPT-5.4-nano	P1-P3	0.00	0.00	100.00
InternVL3.5-8B	P1-P3	0.00	0.00	100.00
InternVL3.5-4B	P1-P3	0.00	0.00	100.00
InternVL3.5-2B	P1-P3	0.00	0.00	100.00
Gemma-3-4B-it	P1-P3	0.00	0.00	100.00

Table 15: Per-prompt strict exact-match classification on the images whose in-image text deviates from standard spelling. Each output is classified as an exact match to the verbatim gold transcription, to the orthographically standard form, or to neither. The seven models that never produce an exact match under any prompt are collapsed to a single P1-P3 row. Rows sum to 100 up to rounding.